



AI Assisted Transport Planning to Enhance the Performance of Transit Oriented Development

**International symposium & workshop
between the University of Seoul and the University of Texas at Austin**

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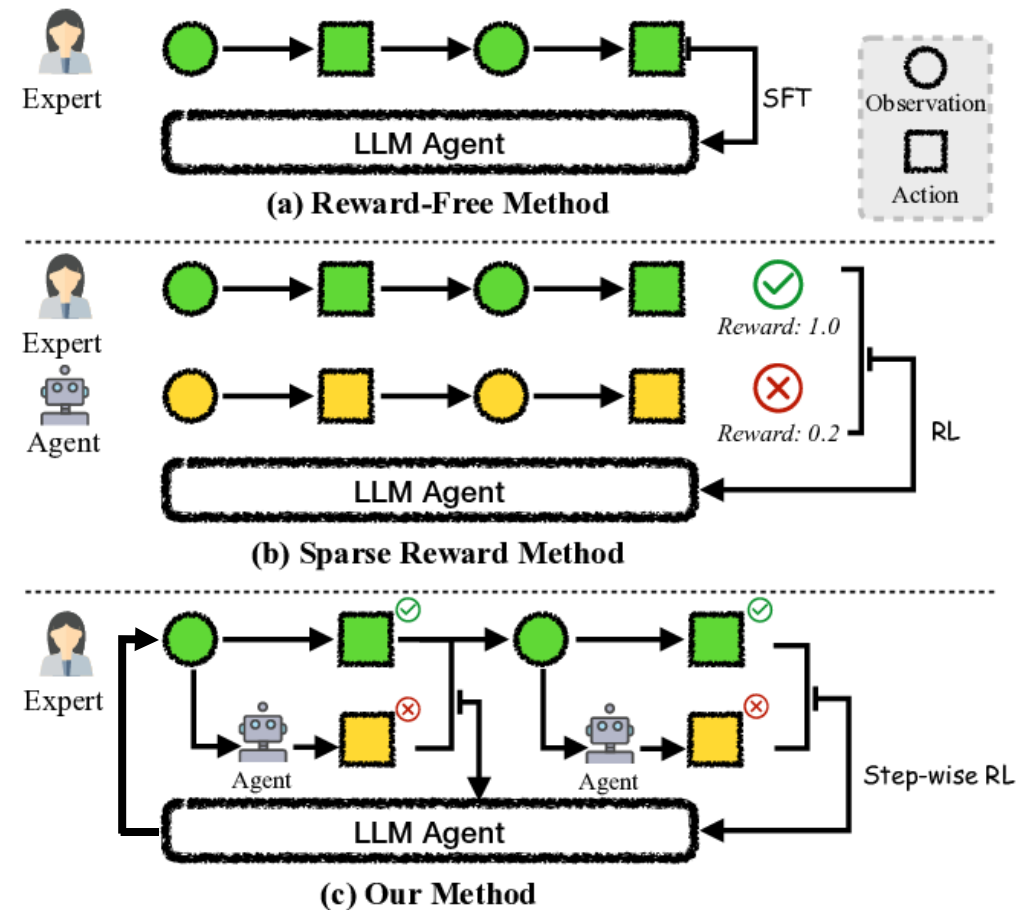
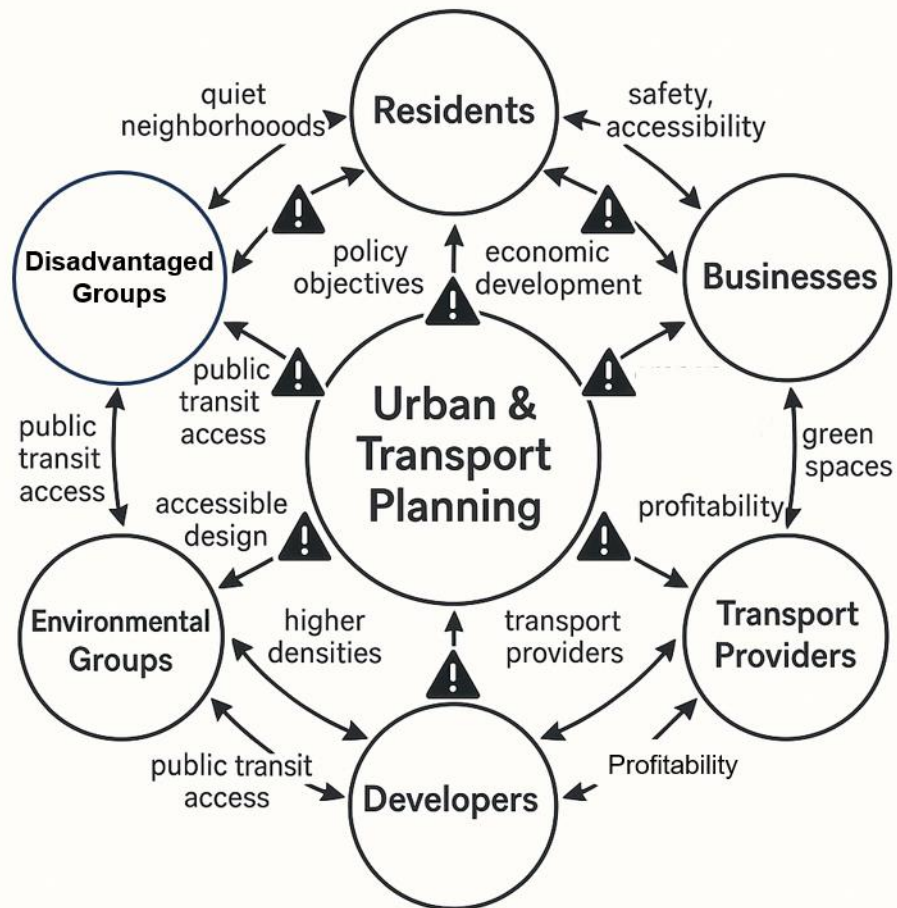
1. Introduction

- I. Motivation**
- II. Academic background**
- III. Research topic and objective**

I Motivation

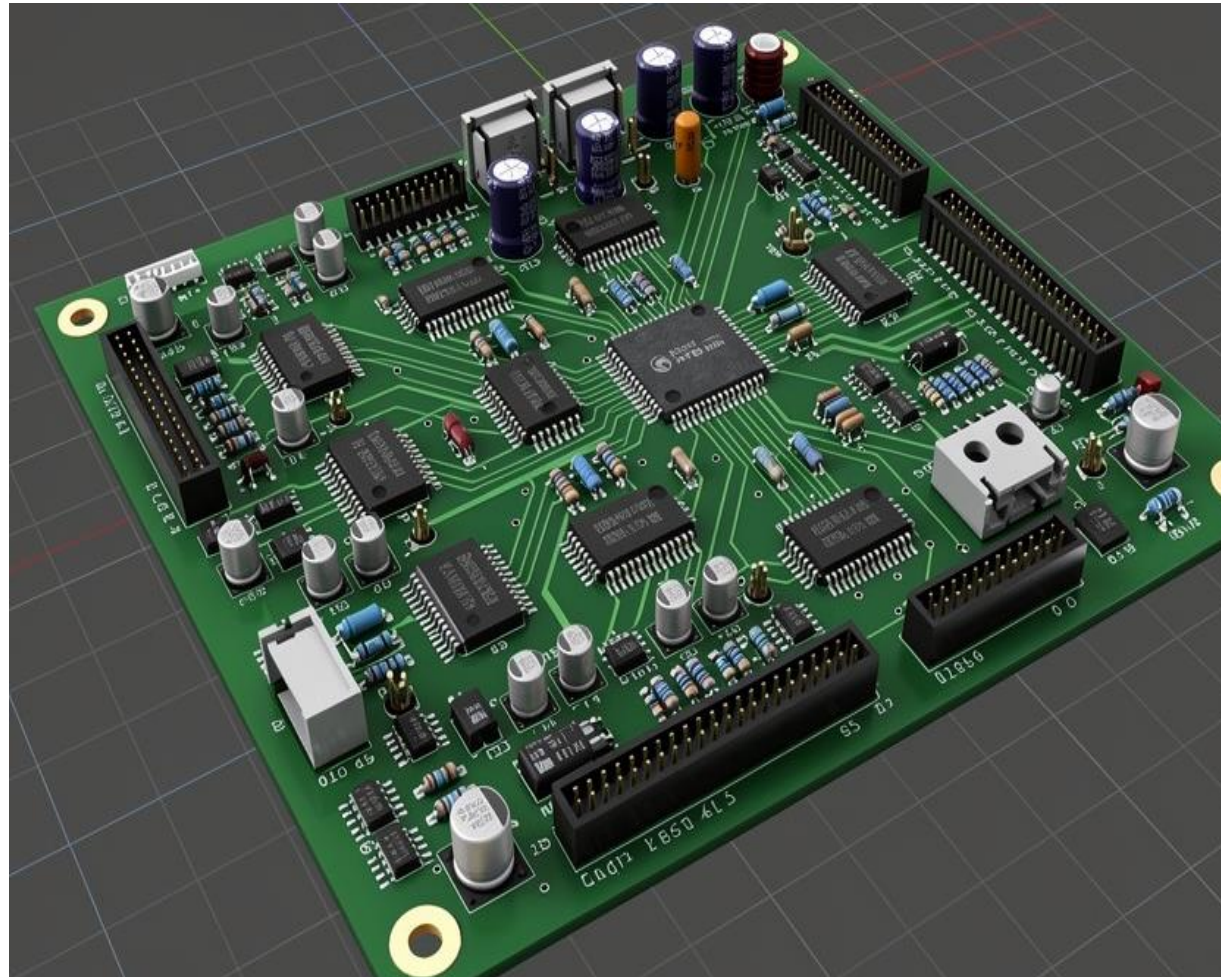
■ The challenges of a planning process and the role of AI

- ✓ The planning process is difficult to accommodate all the needs because the needs of various stakeholders are intertwined
- ✓ LLM can easily and quickly feed back existing decisions to improve



I Motivation

■ PCB(Printed Circuit Board) layout design



I Motivation

■ PCB layout design with AI

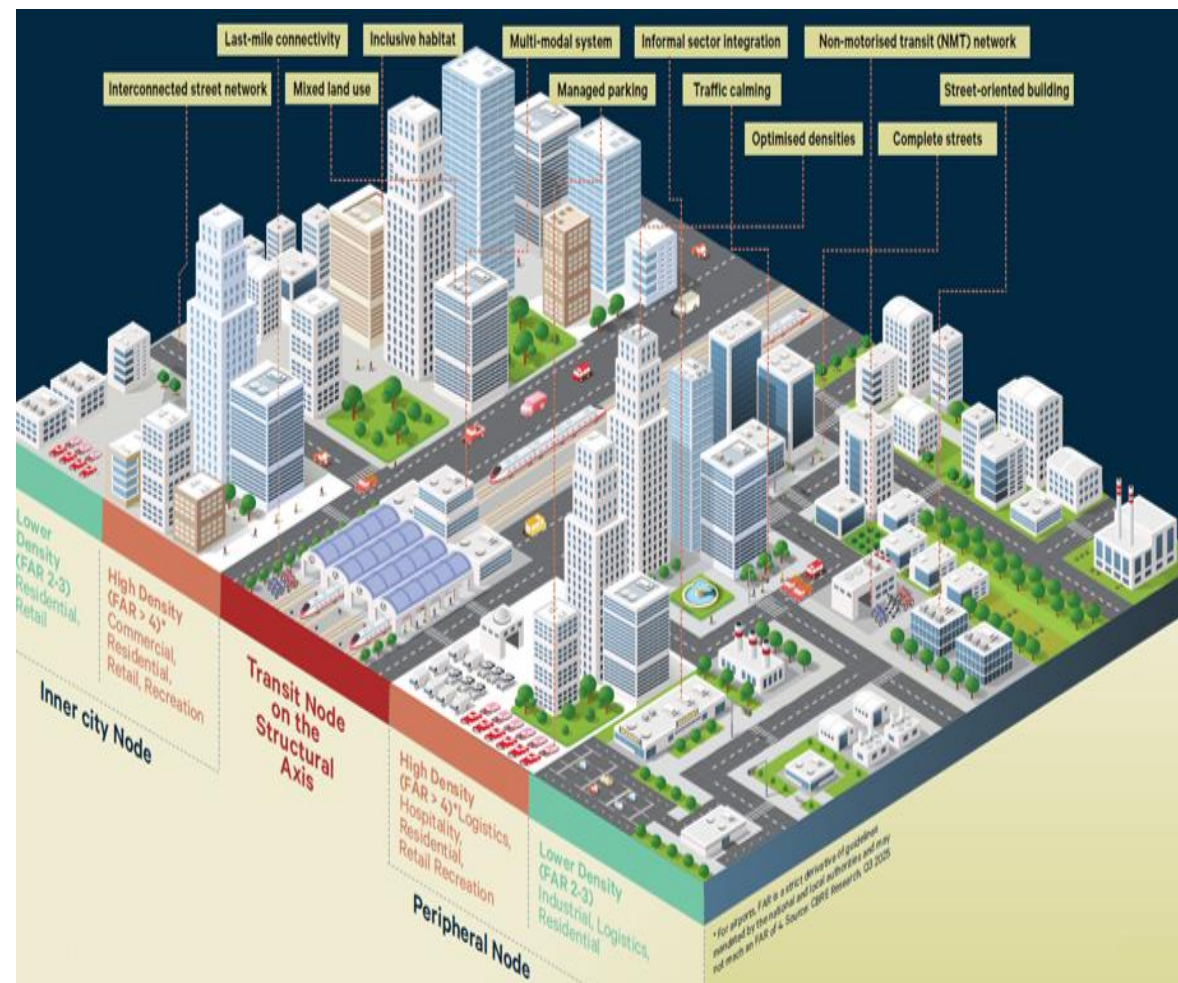
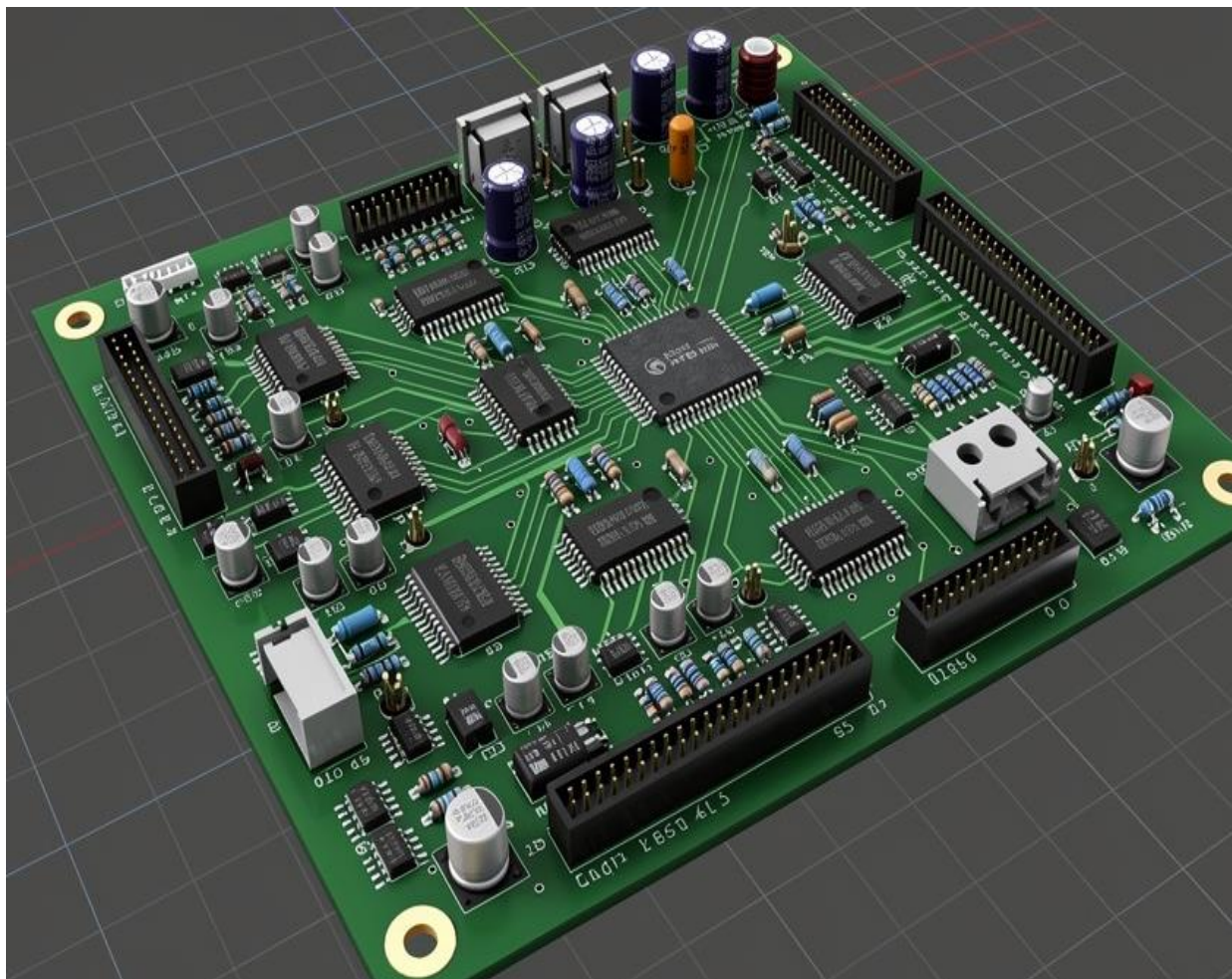


I Motivation

■ PCB layout design



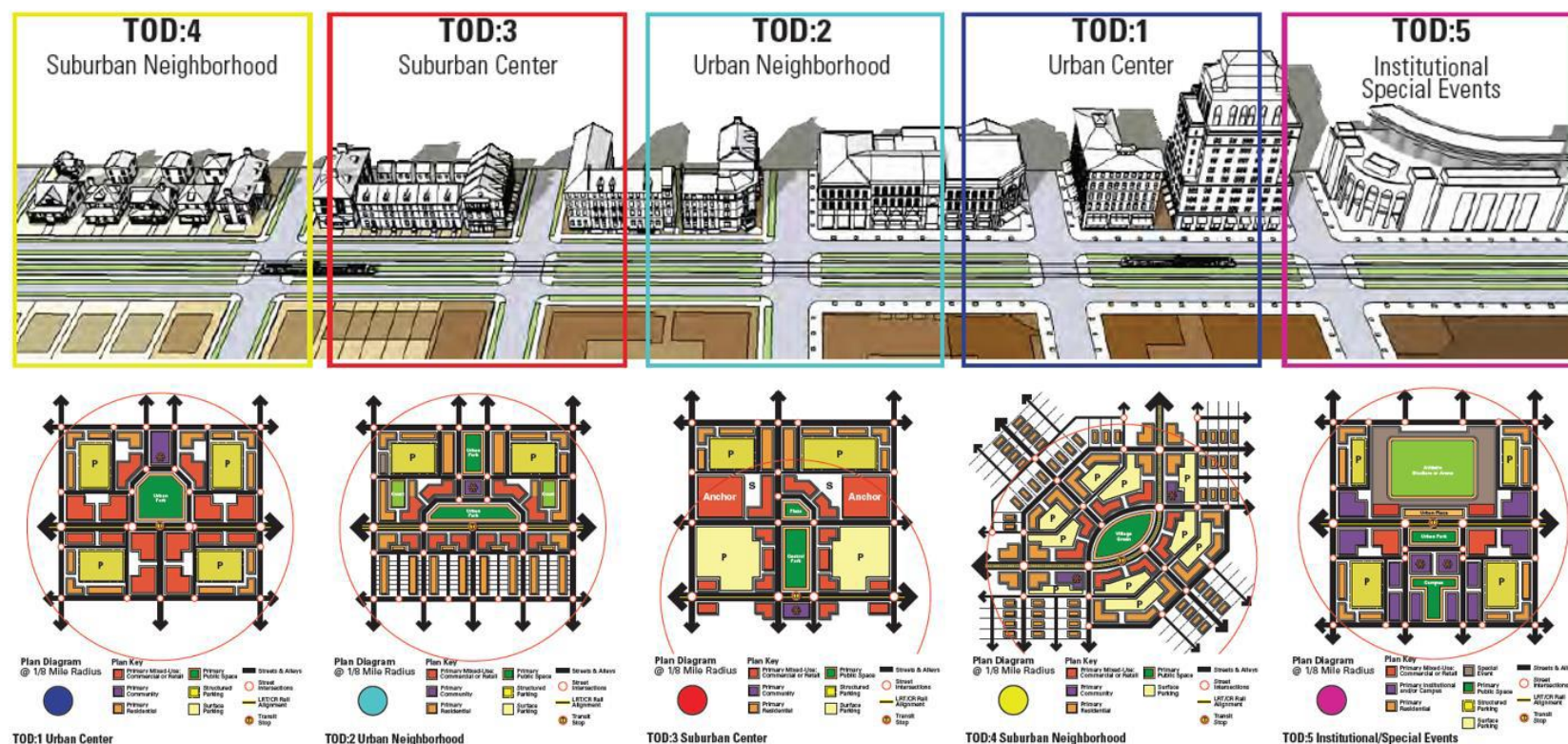
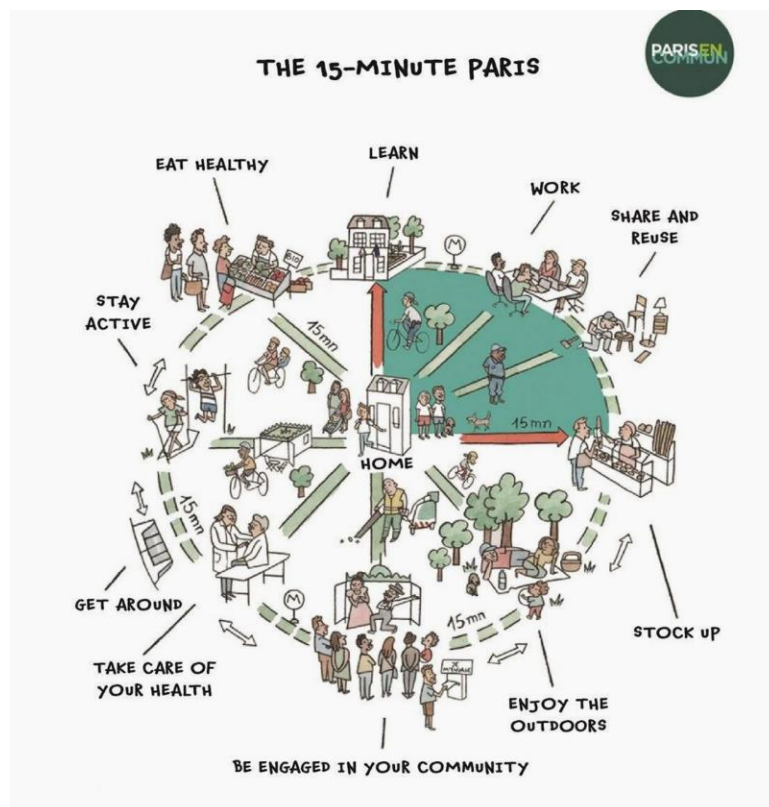
City layout design



II Academic background

■ Transit-Oriented Development(TOD)

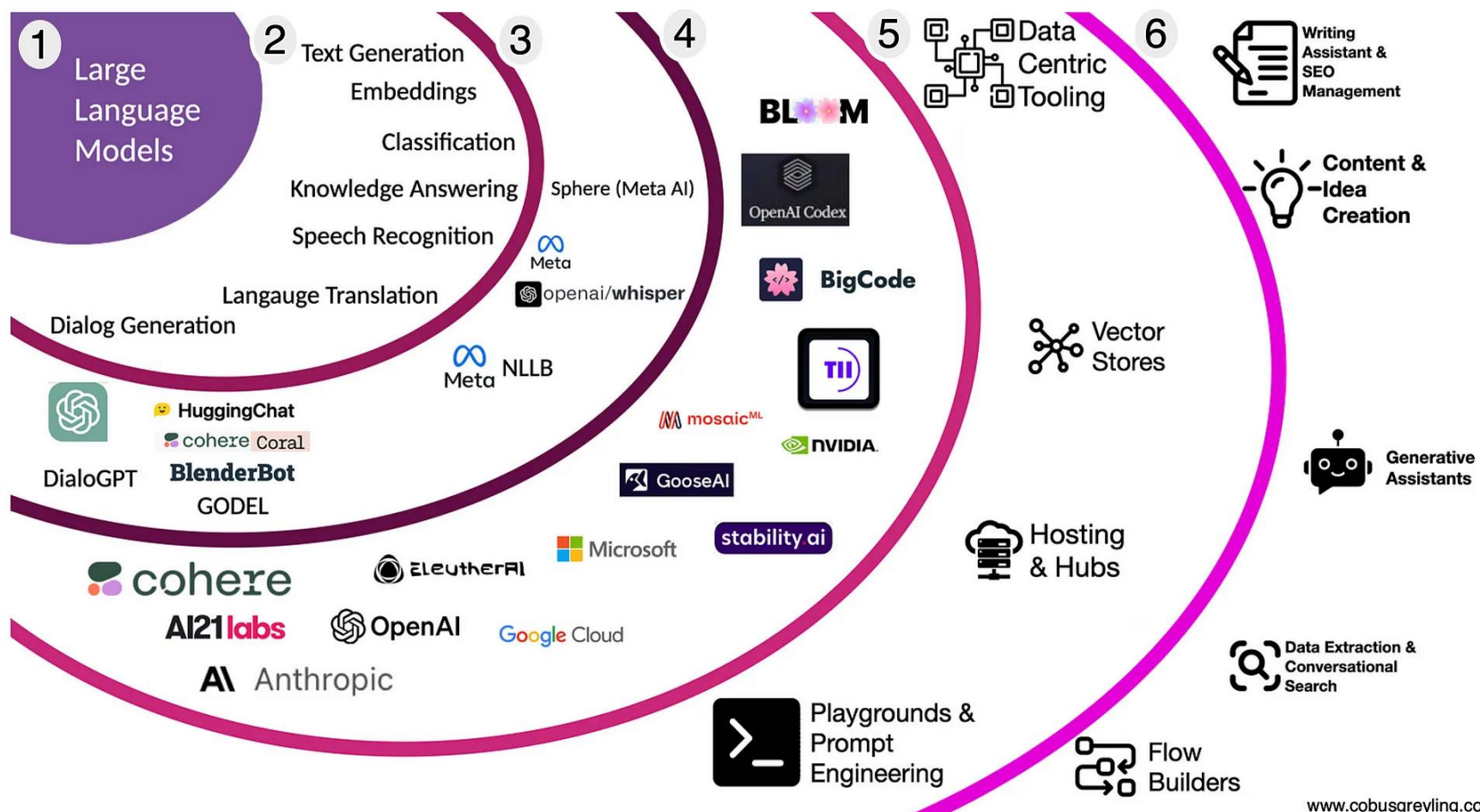
- ✓ Paris's **15-minute city concept** is in the spotlight as a key solution to urban planning.
- ✓ TOD is a type of urban development that maximizes the amount of residential, business and leisure space within walking distance of public transport.
- ✓ TOD in the context of the **3Ds of the built environment: Density, Design and Diversity** (Cervero and Kockelman, 1997)



II Academic background

Large Language Models (LLMs)

- ✓ LLMs can **easily and quickly** generate a variety of solutions.



- ✓ 1 - Available Large Language Models
- ✓ 2 - General Use-Cases
- ✓ 3 - Specific Implementations
- ✓ 4 - Models
- ✓ 5 - Foundation Tooling
- ✓ 6 - End User UIs

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III Research topic and objective

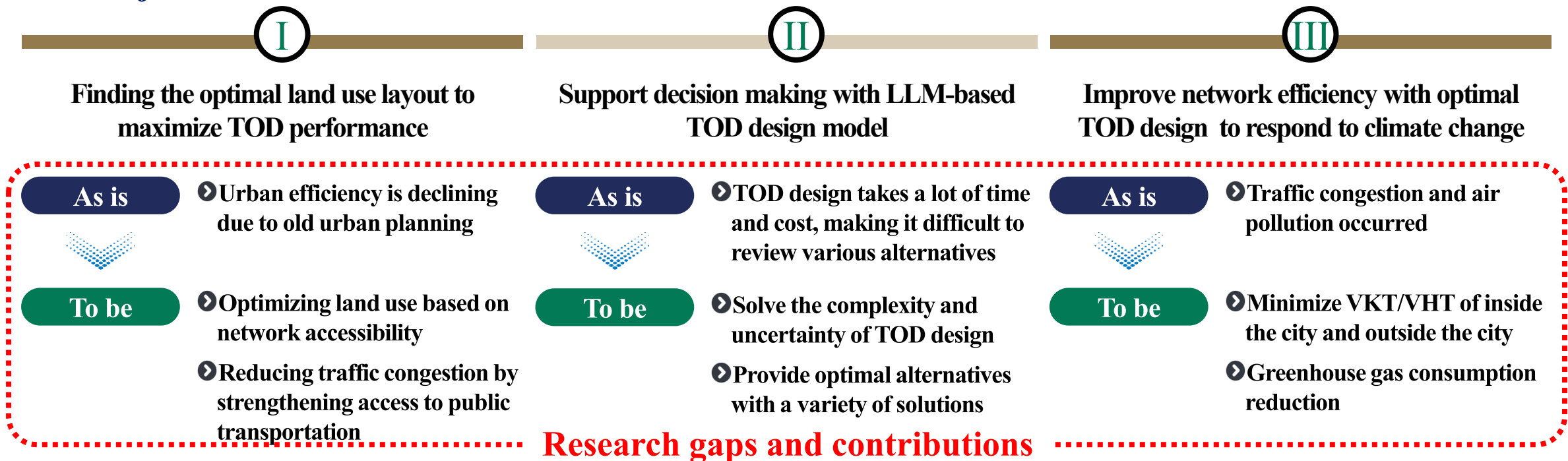
■ Topic

- ✓ Finding the optimal land use layout to maximize TOD performance through a combination of land use type and floor area ratio

■ Hypothesis

- ✓ AI-assisted TOD planning can significantly improve transit ridership and overall planning performance.

■ Objective



2. Research Design

- I. Literature review**
- II. Framework of the study**
- III. Methodology**
- IV. Data description**

I Literature review

■ Transit-Oriented Development(TOD)

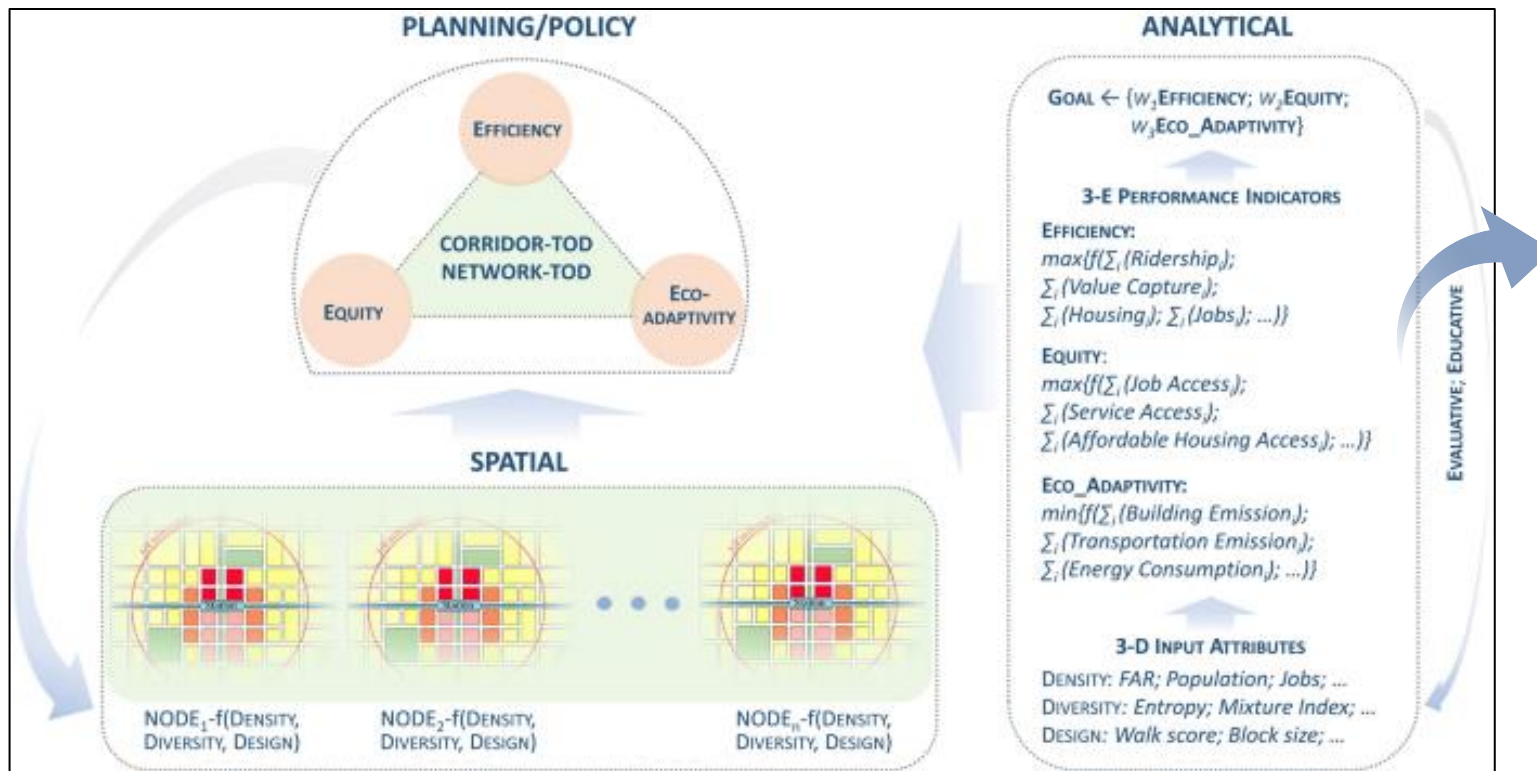
- ✓ TOD is a type of urban development that maximizes the amount of residential, business and leisure space within walking distance of public transport.
- ✓ However, there are **various side effects in terms of transportation and social aspects**.

Authors	Year	Main Contents
Carlton	2009	The concept Transit-oriented Development (TOD) emerged in the context of the New Urbanism movement in the USA in the late 20th century, in a publication on the ecology of communities
Bernick and Cervero	1997	Some of the main benefits of TOD would be to reduce the dependency on motor vehicles, achieved through a combination of mixed communities and proximity between people, activities and transport services
Loo and du Verle	2017	Transit-Oriented Development (TOD) has been proposed by several authors as an effective alternative for the promotion of urban sustainability
Cervero	2014	The integration between transport and land use policies is seen as key ingredient for promoting accessibility, active mobility, and public transport
Turbay, et al.	2022	Curitiba's TOD has had limited influence on population densities, but contributed for the unequal distribution of its benefits, privileging high-income classes and premium real-estate along its main BRT corridors.
C. E. Jones	2023	There are TOD-induced gentrification and displacement risk
F. R. Ashik et al.	2022	There are congestion, emission concentration, and heat island effects in TOD-areas

I Literature review

■ Transit-Oriented Development(TOD)

- ✓ The combination of land use and floor area ratio is tracked to improve 3-E performance indicators



Division	Variables
Density	Floor Area Ratio(FAR) Population
Diversity	Type of Land use Land use Mix
Design	Catchment area Block size

Division	Variables
Efficiency	Transit ridership
Equity	Accessibility
Eco adaptivity	Transportation Emission

I Literature review

■ Land use and transport model

- ✓ Various land-use transport models have been developed, but they are **not fully working** due to the large size of the zone or the problem of inter-variable complexity.
- ✓ Because the change in land use is very slow compared to the change in transportation, there is a problem with the reality of modeling.

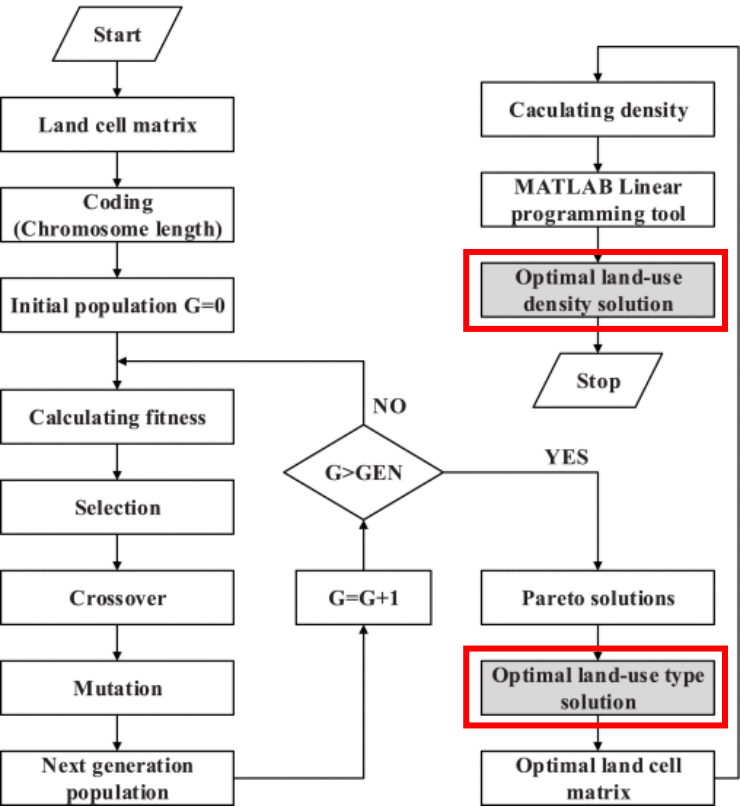
Model	Short description
BOYCE	• the combined models of location and travel choice developed by Boyce
DELTA	• the land-use/economic modelling package by Davids Simmonds Consultancy, Cambridge, UK
ILUTE	• the Integrated Land Use, Transportation, Environment modelling system under development at several Canadian universities
IMREL	• the Integrated Model of Residential and Employment Location developed at the Royal Institute of Technology, Stockholm
ITLUP	• the Integrated Transportation and Land Use Package consisting of the residential location model and the employment model
KIM	• the non-linear urban equilibrium model developed at the University of Illinois at Urbana
MUSSA	• the '5-Stage Land-Use Transport Model' developed for Santiago de Chile by Martinez
PECAS	• the Production, Exchange and Consumption Allocation System developed at the University of Calgary
POLIS	• the Projective Optimization Land Use Information System developed by Prastacos for the Association of Bay Area Governments
STASA	• the master-equation based transport and urban/regional model developed for the metropolitan region of Stuttgart
URBANSIM	• the microeconomic model of location choice of households and firms

I Literature review

■ limitations of similar research

- ✓ Land-use layout optimization and density optimization were performed separately.
- ✓ The transit ridership is calculated as a product of simple parameters

Flowchart of solution approach



The study area



Equation: Maximization of URT Ridership

$$\max Z_2 = \vartheta \cdot \omega \left\{ \sum_{(i,j) \in Z} \sum_{k=1,2,3} \tau^k \left[S \cdot X_{ij}^k \cdot r_{ij}^k (\delta_a^k + \delta_g^k) \right] \right\}$$

Variables		Definition		
i, j	Cell number of row and column in study area, $1 \leq i, j \leq 30$			
Z	Zone division in study scope, $Z=\{Z_1, Z_2, Z_3\}$, Z_1 represent internal zone, Z_2 represent intermediate zone, Z_3 represent external zone			
k	Types of land-use, which is coded in this paper as follows: commercial and office as 1; residential as 2; public space as 3; green space as 4; transport facilities as 5; pending to develop as 0; out of the study scope as 9			
X_{ij}^k	Land-use type index of cell (i, j) for land-use k , which is a binary variable that equals 1 if land-use type of cell (i, j) is k and equals 0 otherwise, $k=\{1, 2, 3\}$			
r_{ij}^k	Land-use density of type k at cell (i, j) , $k=\{1, 2, 3\}$			
S	Area of cell (i, j) , $60\text{m} \times 60\text{m} = 3600\text{m}^2$			

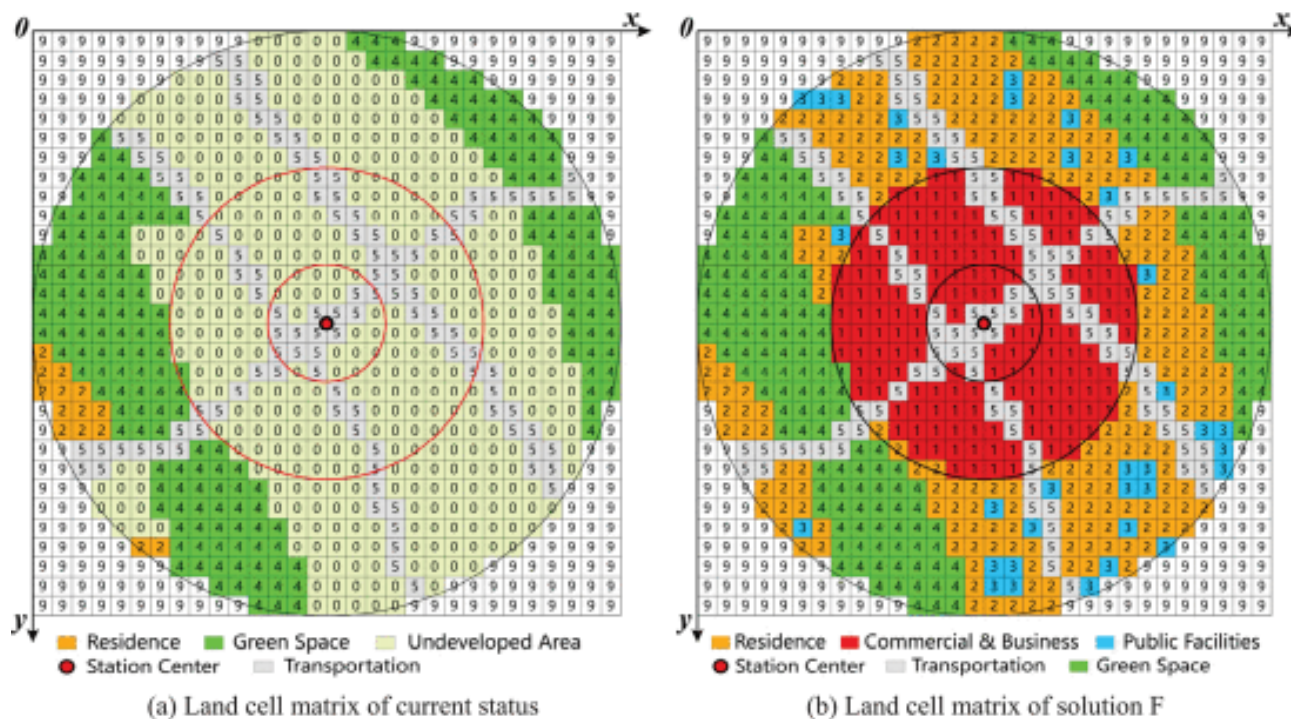
Symbol	Definition	Value		Notes
		Commercial type station	Residential type station	
δ_a^k	Trip attraction rate of cell for land-use type k	817 person/10,000 m ²		$k=1$
		171 person/10,000 m ²		$k=2$
		250 person/10,000 m ²		$k=3$
δ_g^k	Trip generation rate of cell for land-use type k	621 person/10,000 m ²		$k=1$
		400 person/10,000 m ²		$k=2$
		200 person/10,000 m ²		$k=3$
r^k	Peak hour coefficient of land-use type k		0.14	$k=1$
			0.12	$k=2$
			0.06	$k=3$
ϑ	External travel coefficient of residents in study scope	0.75	0.93	
ω	Urban rail transit mode share rate		0.472	

I Literature review

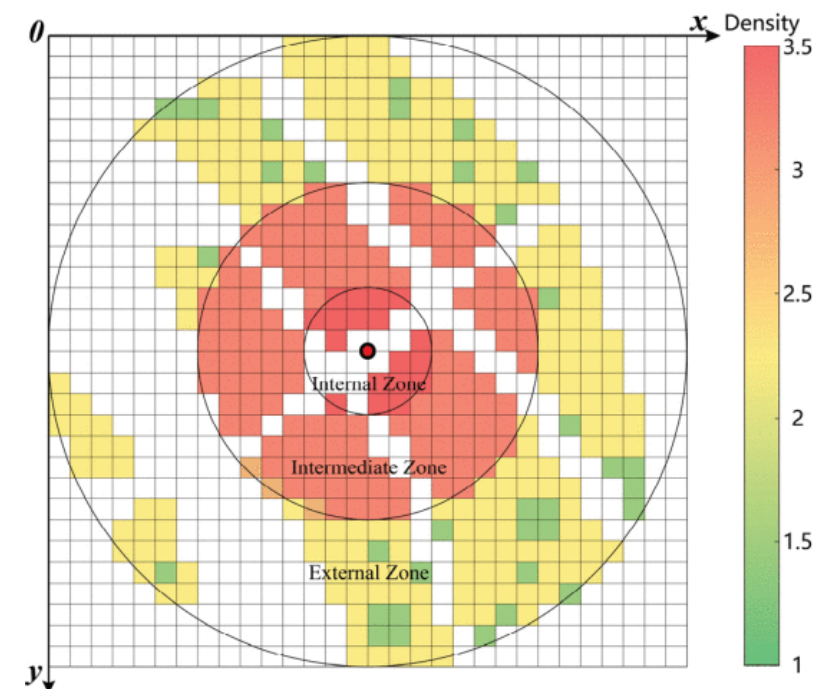
■ limitations of similar research

- ✓ Some studies failed to consider the network-based practical TOD area.
- ✓ As a result, centralized land use and dense results have been derived.

Land cell matrix of Sidaoqiao station



Optimization result of land development density



I Literature review

■ Large Language Models(LLMs) for urban planning

- ✓ In urban studies including planning and geography, interest in LLM applications is growing.
- ✓ While emerging LLMs show promise in specific tasks like **information extraction and tool interaction**, their potential to support the entire planning process and foster dialogue among stakeholders has not been fully explored.

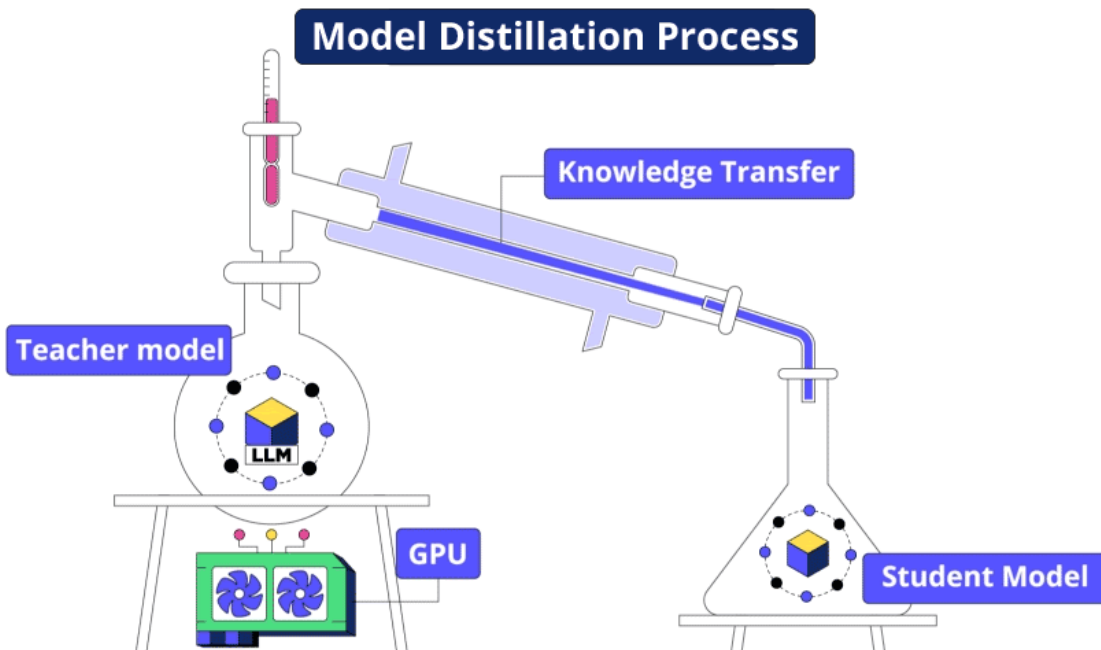
Authors	Year	Main Contents
Wang et al.	2024	A recent literature review categorized LLM and generative AI use in geography into four areas: map generation, geotechnical engineering, spatial analysis, and transportation.
Rane	2023	ChatGPT's roles and challenges in urban planning were highlighted, focusing on its applications in data analysis, prediction, design, visualization, citizen engagement, and policymaking.
Daniel	2023	The American Planning Association's Planning Advisory Service (PAS) explored ChatGPT’s potential in automating planning tasks like report writing, coding, and workflow creation, though concerns remain over output reliability, data formatting, infrastructure, and disinformation in public engagement.
Fu, Sanchez, et al.	2024	LLMs have also been used to analyze public feedback, such as summarizing responses to urban densification issue.
Y. Zhang et al.	2024	GeoGPT was developed, combining LLMs with GIS tools for tasks like spatial queries and mapping.
Li et al.	2024	UrbanGPT was created for spatiotemporal predictions, outperforming traditional methods.
S. Zhang et al.	2024	TrafficGPT was introduced, and it uses LLMs to manage big traffic data analysis and display, offering a user-friendly way to select models and provide feedback.
Peng et al.	2024	Most studies focus on specific tasks within planning, with little attention given to integrating LLMs into the broader planning process, a central issue in current debates on AI in planning.

I Literature review

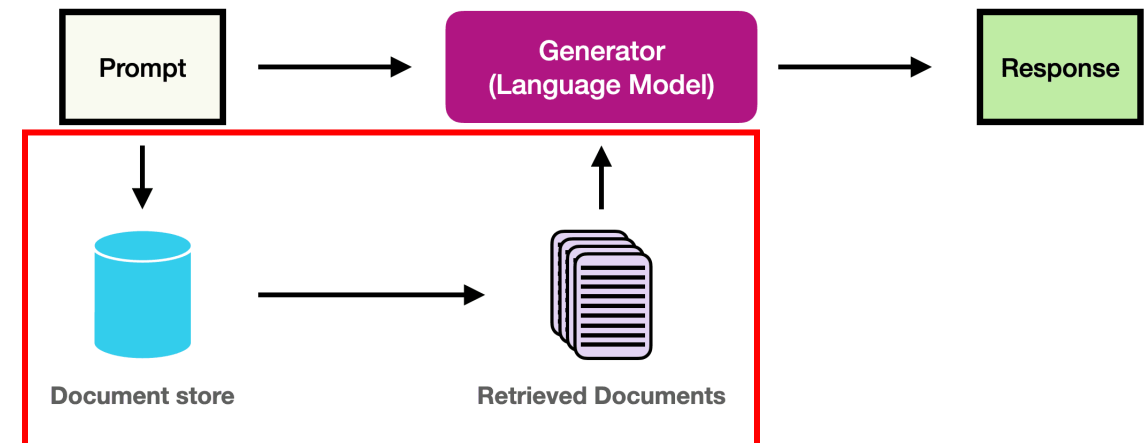
■ Large Language Models(LLMs) for TOD design with AI distillation

- ✓ Use existing open API to build models with AI distilling
- ✓ Strengthen AI expertise based on Retrieval Augmented Generation(RAG)

AI distillation



Retrieval Augmented Generation(RAG)



II Framework of the study

■ Transit-Oriented Development Design

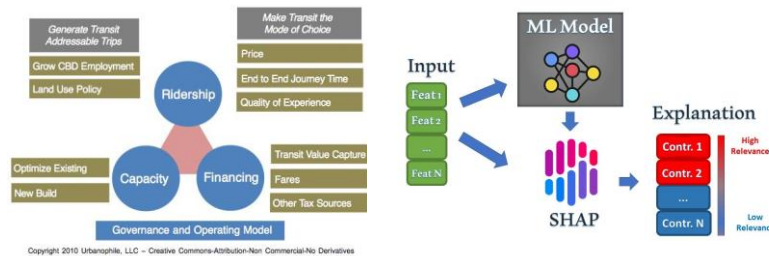
- ✓ This study integrates the transit ridership estimation, scenario generation, and genetic algorithm-based optimization into a LLM



Large Language Model

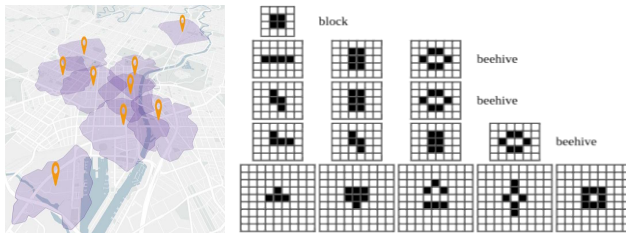
1) Transit ridership model with AI

- Based on AI, build a transit ridership model
- The transit ridership model estimates the change in transit ridership rate

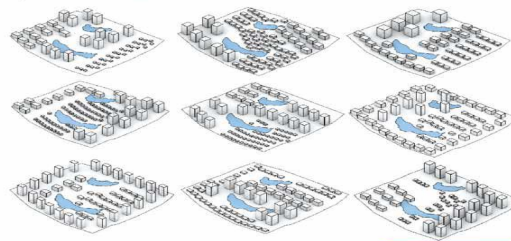


2) Land use scenario generation

- Set up the catchment area and land use scenarios by cells

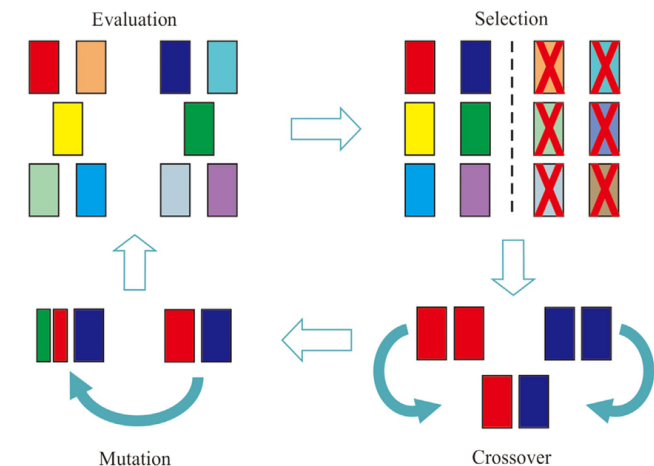


- Land use scenarios generation



3) Optimizing with a genetic algorithm

- Select optimal alternatives to maximize transit ridership



Maximize 3-E performance indicators

Transit ridership model with AI

- ✓ Building and validating the Transit Ridership model based on XAI
- ✓ Compare the existing 4-step model, econometric model, and AI model to calculate the modal-sharing ratio

Incremental logit model

$$P_i^* = \frac{P_i \cdot \exp \Delta V_i}{\sum_j P_j \cdot \exp \Delta V_j}$$

여기서 P_i^* : 사업 시행시 수단 i 의 선택확률
 P_i : 사업 미시행시 수단 i 의 관측분담률
 ΔV_i : 사업 시행 전·후 수단 i 의 효용변화

승용차 효용 = $\beta_{pr} \times Time + \gamma_{pr} \times Acost$

택시 효용 = $\alpha_T + \beta_{pr} \times Time + \gamma_{pr} \times Tcost$

버스 효용 = $\alpha_B + \beta_{1,ph} \times Vtime + \beta_{2,ph} \times Outtime + \gamma_{ph} \times Bcost$

전철 효용 = $\alpha_S + \beta_{1,ph} \times Vtime - \beta_{2,ph} \times Outtime - \gamma_{ph} \times Scost$

버스+전철 효용 = $\alpha_{BS} + \beta_{1,ph} \times Vtime + \beta_{2,ph} \times Vtime + \gamma_{ph} \times BScost$

여기서, $Time$: 가·종점간 총 통행시간(분)

$Vtime$: 차내통행시간(분)

$Outtime$: 차외통행시간(분)

β_{pr} : 개인교통수단 시간변수의 계수

β_{ph} : 대중교통수단 시간변수의 계수

γ_{pr} : 개인교통수단 비용변수의 계수

γ_{ph} : 대중교통수단 비용변수의 계수

$Acost$: 승용차 통행비용(100원)

$Tcost$: 택시 통행비용(100원)

$Bcost$: 버스 통행비용(100원)

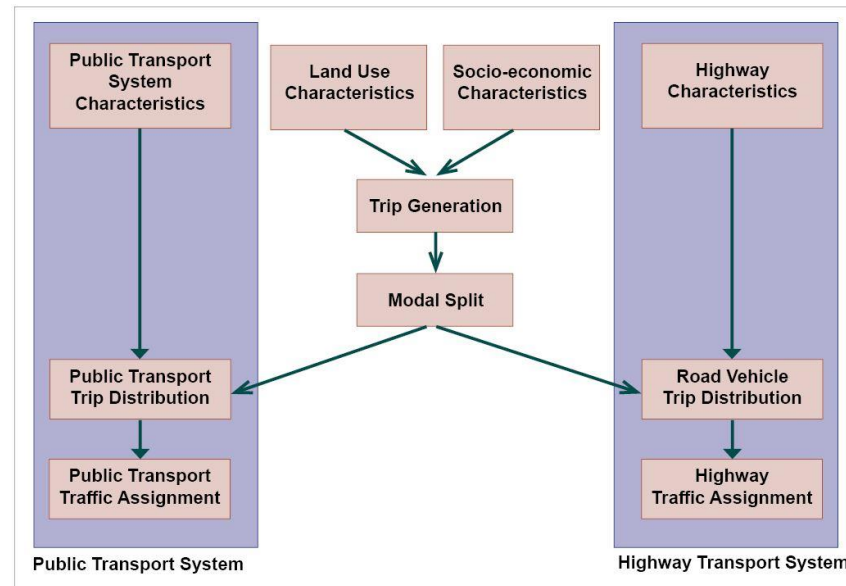
$Scost$: 지하철 통행비용(100원)

$BScost$: 버스+지하철 통행비용(100원)

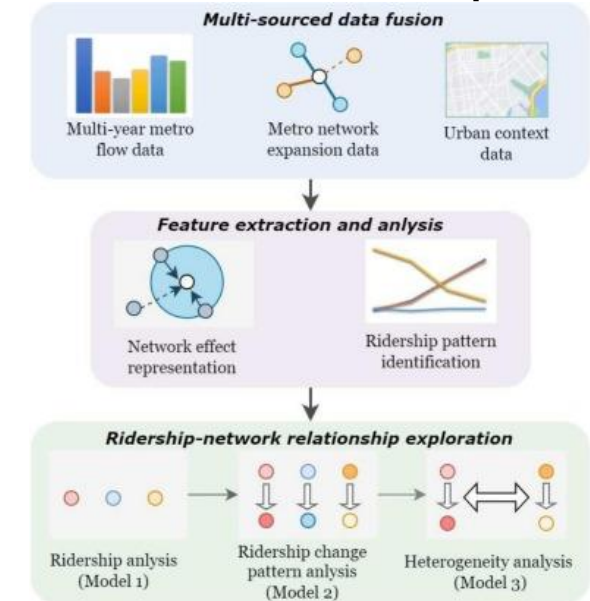
α_m : m 수단의 상수항

자료 : 경기연구원, 「수단선택 및 통행배정 모형(수정)」, 2019.

Econometric model



AI-based transit ridership model

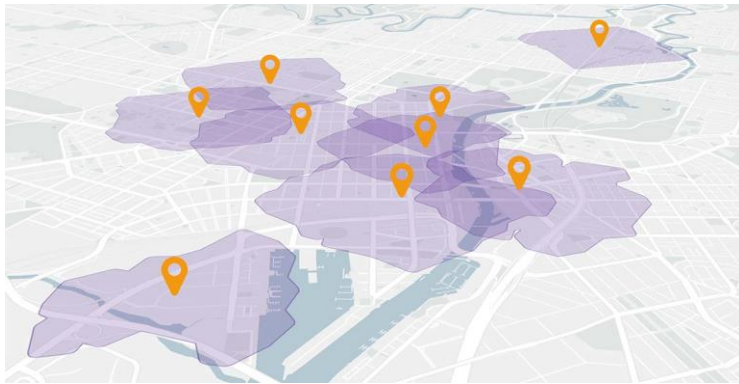


Division	Travel Attributes		Socio-economic Variables		Land use characteristic	
	Time	Distance	Income	population	Land use	FAR
4-step model	O	O	X	X	X	X
Econometric model	O	O	O	O	△	X
AI-based model	O	O	O	O	O	O

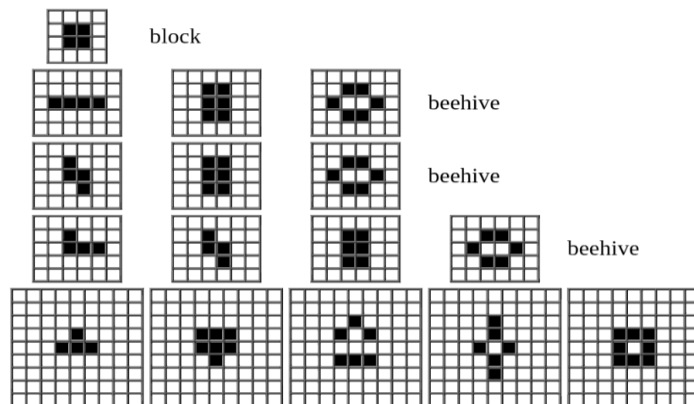
■ Land use scenario generation with Generative model

- ✓ Set the Catchment area based on the network distance and separate the area into cells
- ✓ Create a land-use scenario for each cell to create an urban planning scenario for TOD

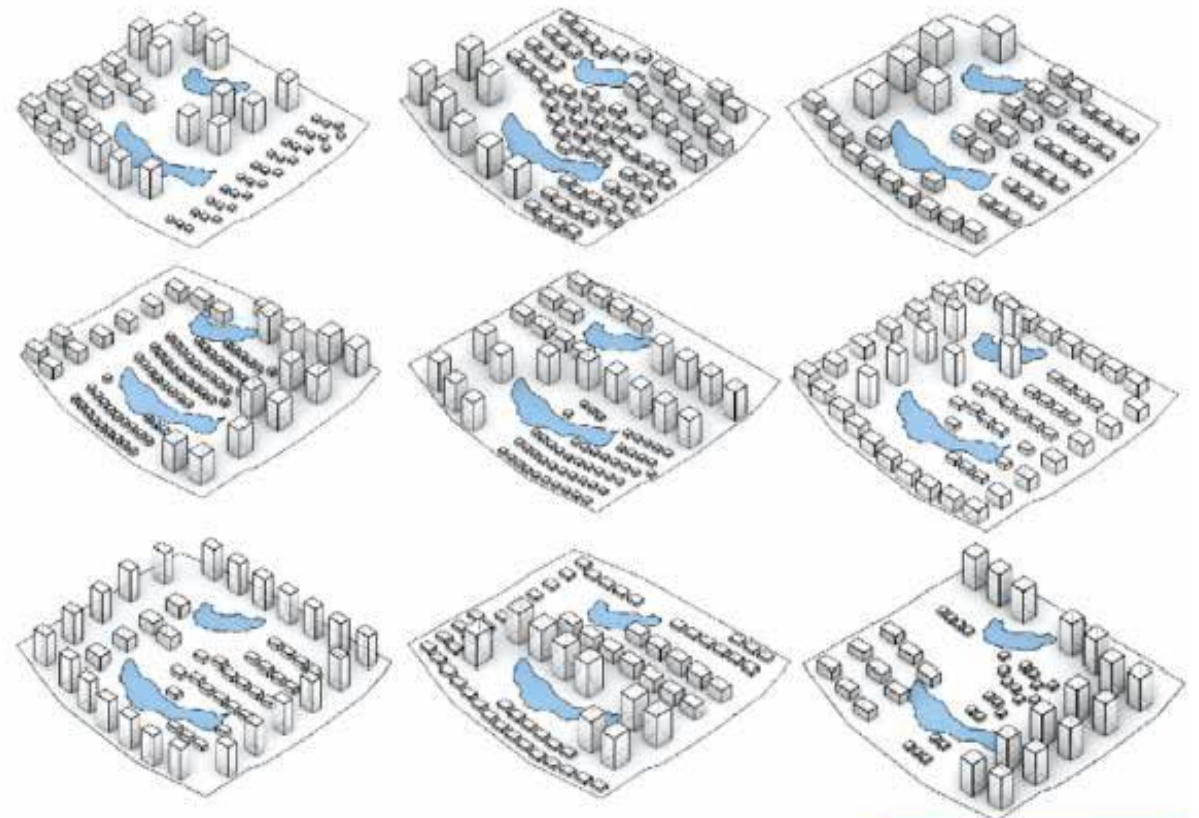
1) Catchment area setting



2) Set up a Land Use scenario by cells

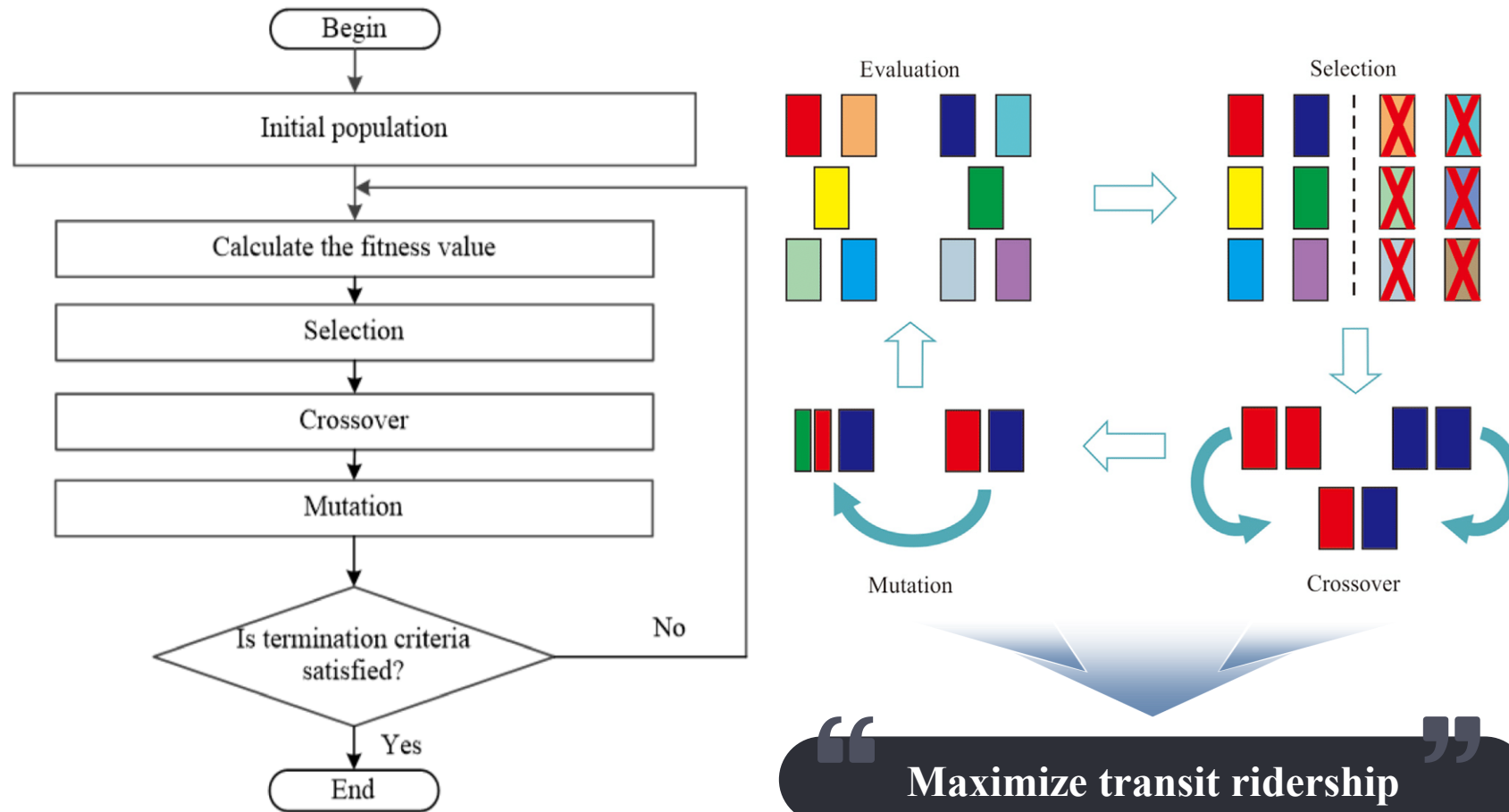


3) Land use scenario generation



■ Optimization with Genetic algorithm

- ✓ Define each scenario created through the previous two steps as Chromosome
- ✓ Optimized by cross-breeding Chromosomes to maximize transit



“Maximize transit ridership”

Effectiveness

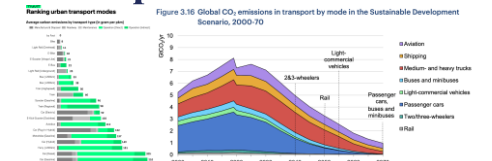
VKT/VHT of inside the city



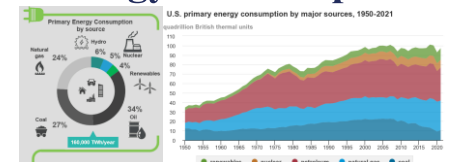
VKT/VHT of outside the city



Transportation Emission

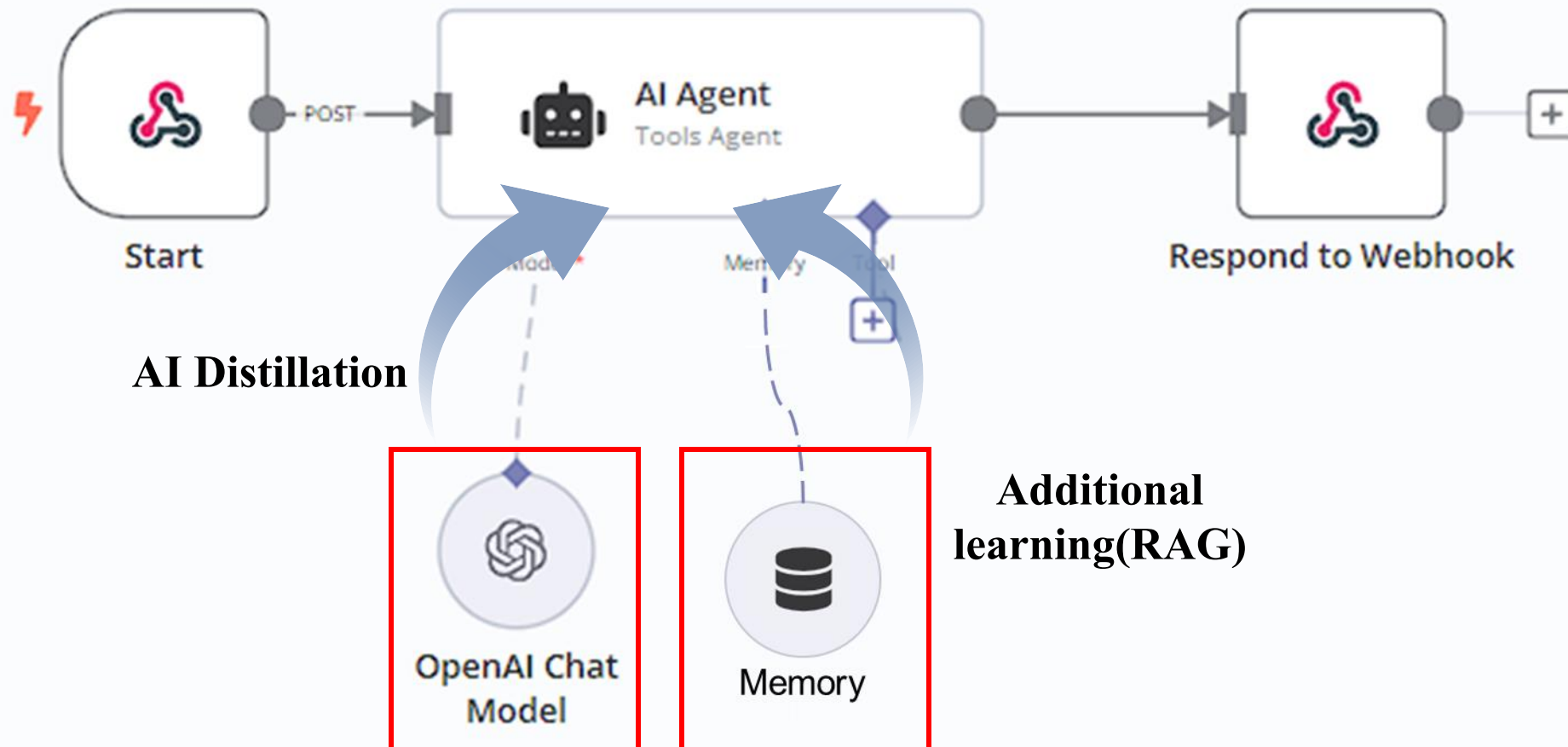


Energy Consumption



■ Integration of the framework with LLM

- ✓ Provide conducted models and data to AI agents in memory form
- ✓ AI agent communicates with users based on memory and open API



IV Data description

■ Land use data

- ✓ Land Use Analysis based on Land Use Status Chart (1:25,000)
- ✓ The land use status is divided into 38 sub-categories and applied to the analysis according to the purpose of this study



Main Category	Subcategory	Detailed Category
Agricultural Land	Paddy Fields	Irrigated Paddy Field Non-Irrigated Paddy Field
	Dry Fields	General and Special Crops Orchards and Others
Forest Land	Grassland	Natural Grassland Artificial Grassland
	Wooded Area	Coniferous Forest Broadleaf Forest Mixed Forest Golf Course
	Other Natural Areas	Recreation Area Cemetery or Memorial Park Rocky Area / Quarry
	Residential and Commercial Areas	General Residential Area High-rise Residential Area Commercial / Business Area Vacant Land / Artificial Green Area
	Transportation Facilities	Road Railroads and Surrounding Areas Airports Ports
Urban and Residential Areas	Industrial Area	Industrial Facilities Vacant Industrial Land and Others
	Public Facilities	Power Generation Facilities Wastewater and Waste Treatment Facilities Educational and Military Facilities Public Land
	Miscellaneous Facilities	Fish Farms / Aquaculture Mining Sites Landfill Sites Hot Springs Livestock Facilities Mudflats Salt Farms
	Water Bodies	River Lake / Reservoir Dam Beaches

Types of land use

Residential area

Office area

Public area

Green area

IV Data description

2. Research design

■ Building data

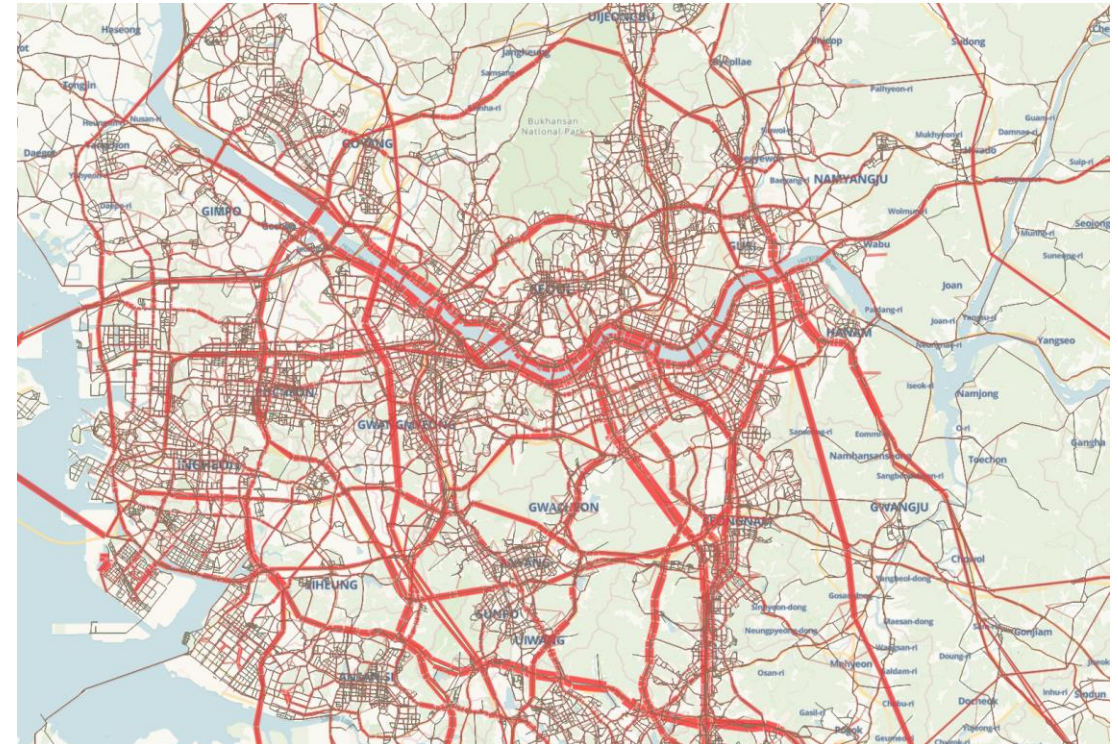
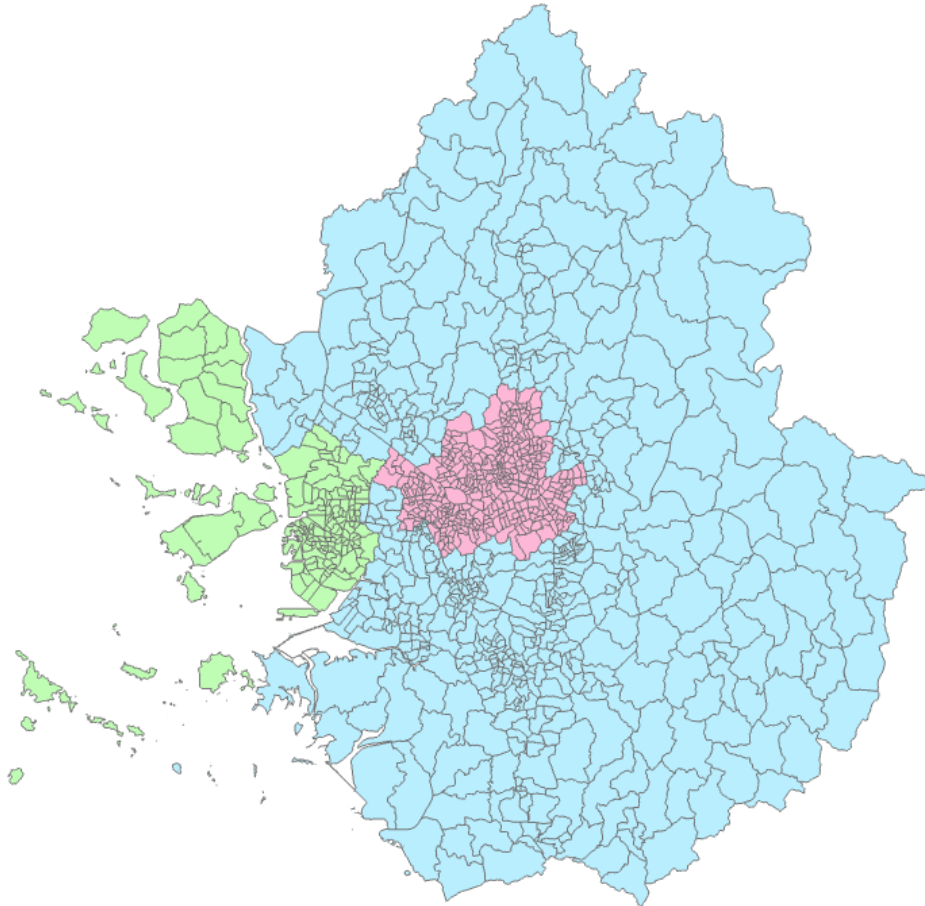
- ✓ Based on GIS building integration information data, it is used for floor area ratio analysis
- ✓ Based on the continuous cadastral information, the spatial information of the building and the property information of the building register of the building administration system are integrated into the building unit



No.	Category Type
1	Source Geometry ID
2	GIS Building Integrated Identifier
3	Unique Identifier
4	Legal District Code
5	Legal District Name
6	Lot Number
7	Special Land Code
8	Special Land Classification Name
9	Building Use Code
10	Building Use Name
11	Building Structure Code
12	Building Structure Name
13	Building Area (㎡)
14	Approval Date for Use
15	Gross Floor Area
16	Site Area (㎡)
17	Height (m)
18	Building Coverage Ratio (%)
19	Floor Area Ratio (%)
20	Building ID
21	Illegal Building Flag
22	Coordinate System Reference Key
23	Data Reference Date
24	Source City/Province/District Code
25	Building Name
26	Building Wing Name
27	Number of Above-Ground Floors
28	Number of Underground Floors
29	Data Creation/Modification Date

■ O/D Trip and Network

- ✓ O/D trip by purpose/mode
- ✓ Road / transit network
- ✓ Spatial scope: 1,135 zones in the Seoul metropolitan area (Seoul, Gyeonggi, Incheon)



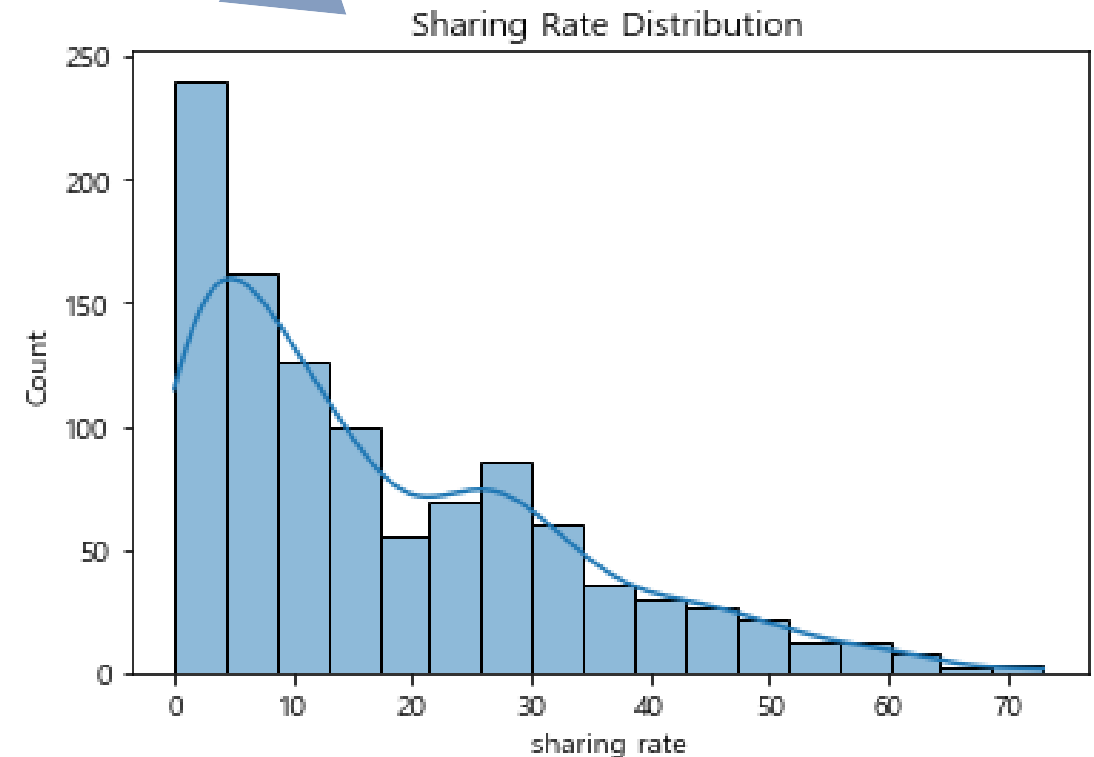
3. Transit ridership modelling with AI

- I. Dataset**
- II. Transit ridership modelling with AI**
- III. Interpretation of relationships based on XAI**

■ Variables setting for modelling

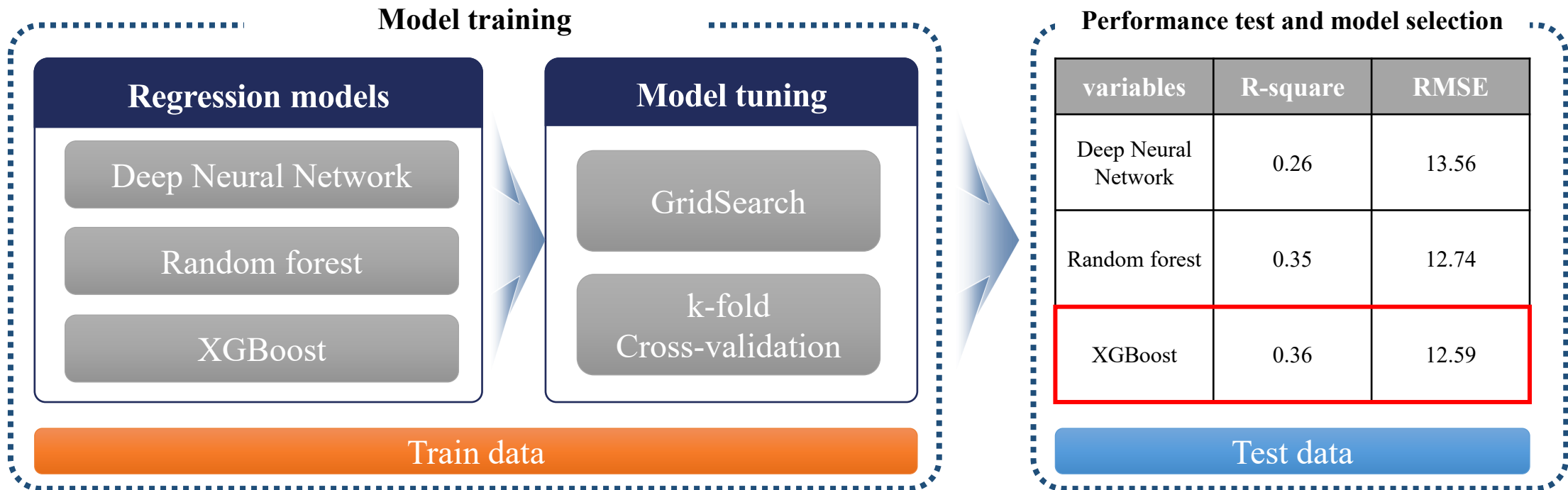
- ✓ The variables necessary for modelling construction were constructed by processing the previous data.

Division		variables	Unit	Sources
Transit ridership ratio		sharing rate	%	OD trip
Land use characteristic	Floor Area Ratio	pub_FAR		Building data
		green_FAR		
		office_FAR		
		res_FAR		
	Land use area	pub_area	km ²	Land use data
		green_area		
		office_area		
		res_area		
Travel Attributes		dis	km	Road network



■ Transit ridership estimation modeling with AI

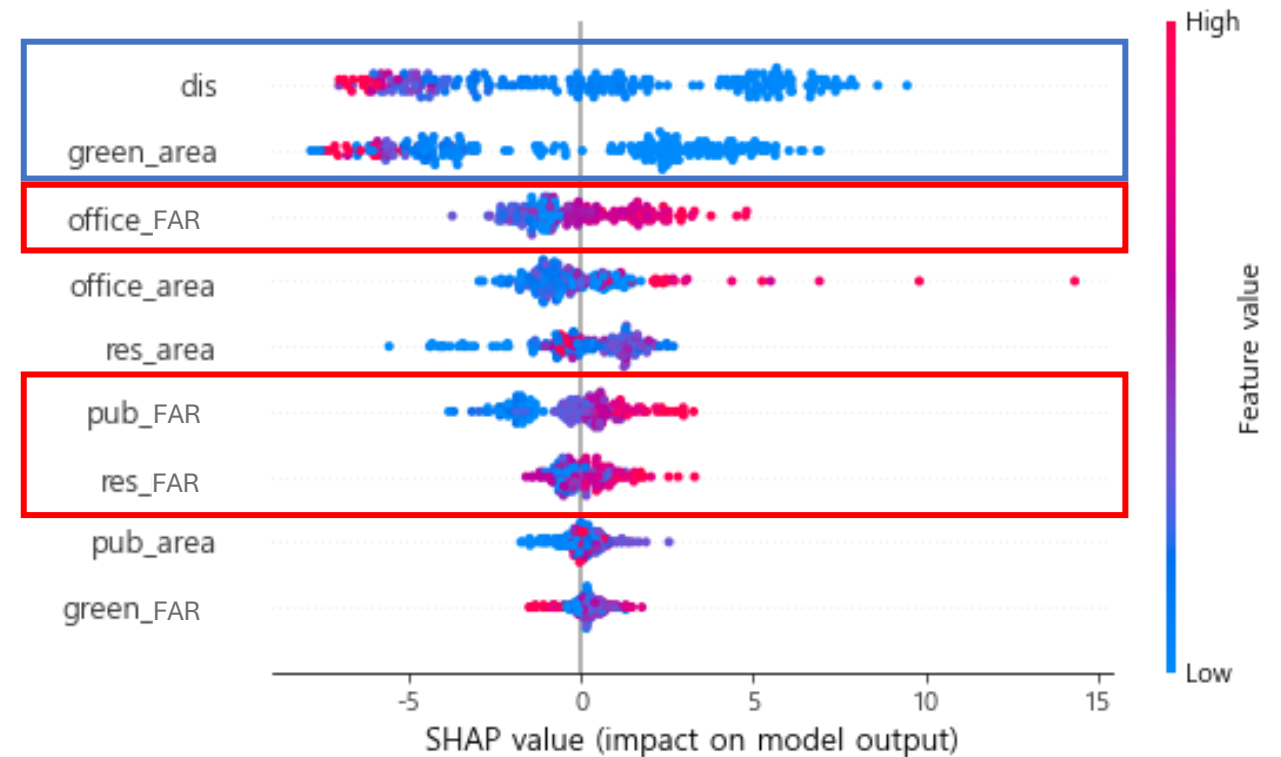
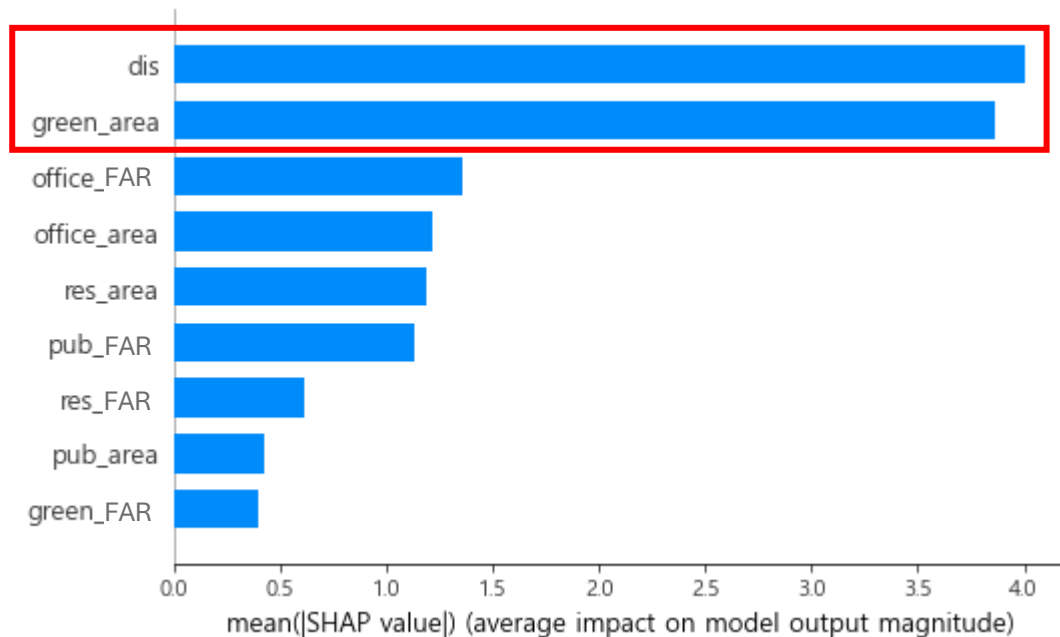
- ✓ The AI model was built, and tuning by grid search and cross-validation method was performed.
- ✓ The XGBoost model showed the highest performance and was selected as the optimal model.



Interpretation of relationships based on XAI

■ Feature importance and SHAP values

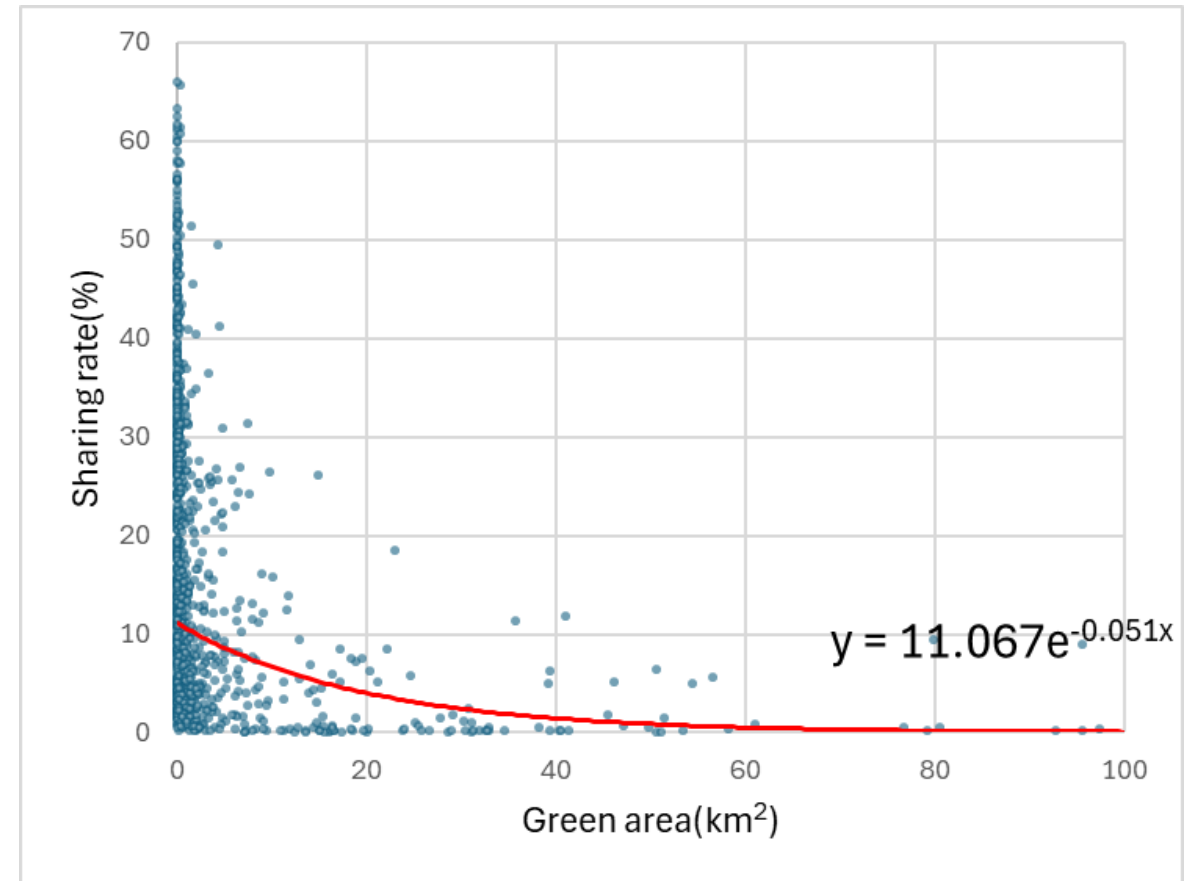
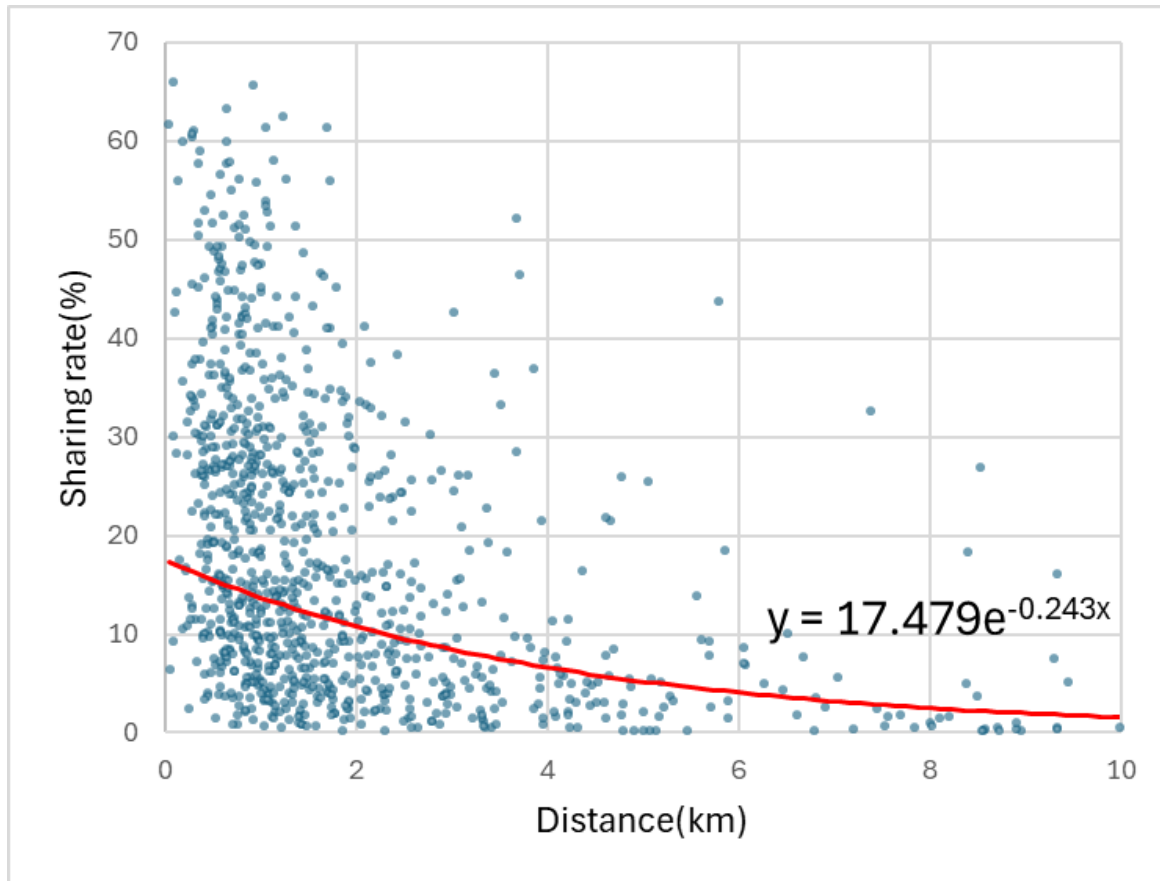
- ✓ The relationship between variables was analyzed using SHAP, the XAI method.
- ✓ The floor area ratio of office, public, and residential areas clearly shows a positive effect on transit ridership



Interpretation of relationships based on XAI

■ Analysis of relationships between key variables (negative effect)

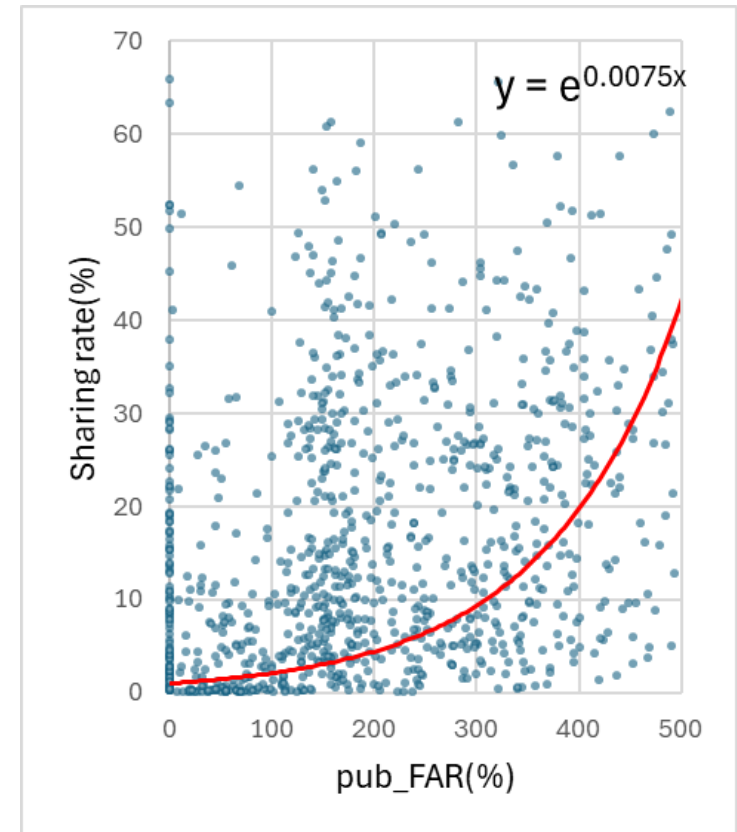
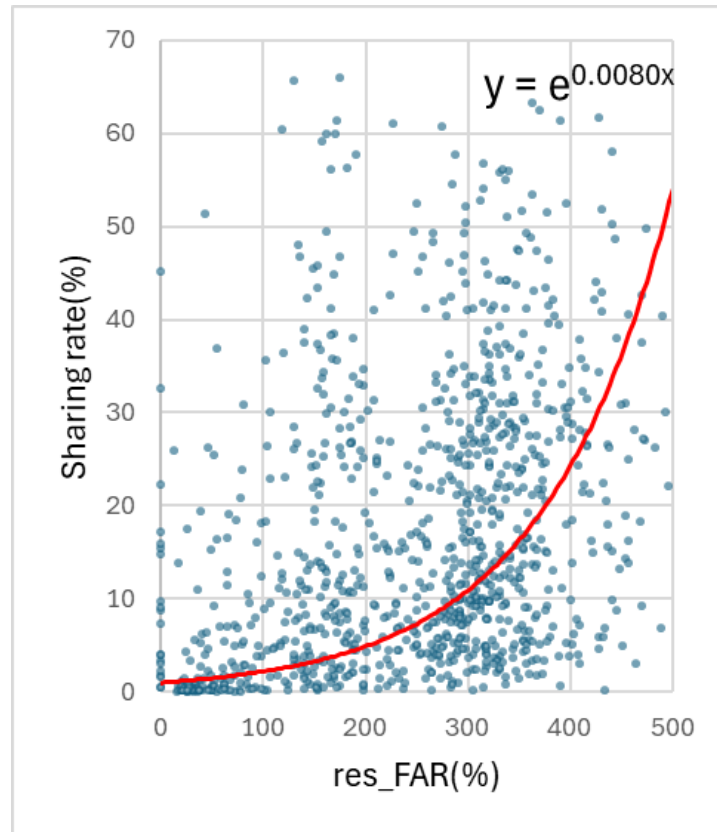
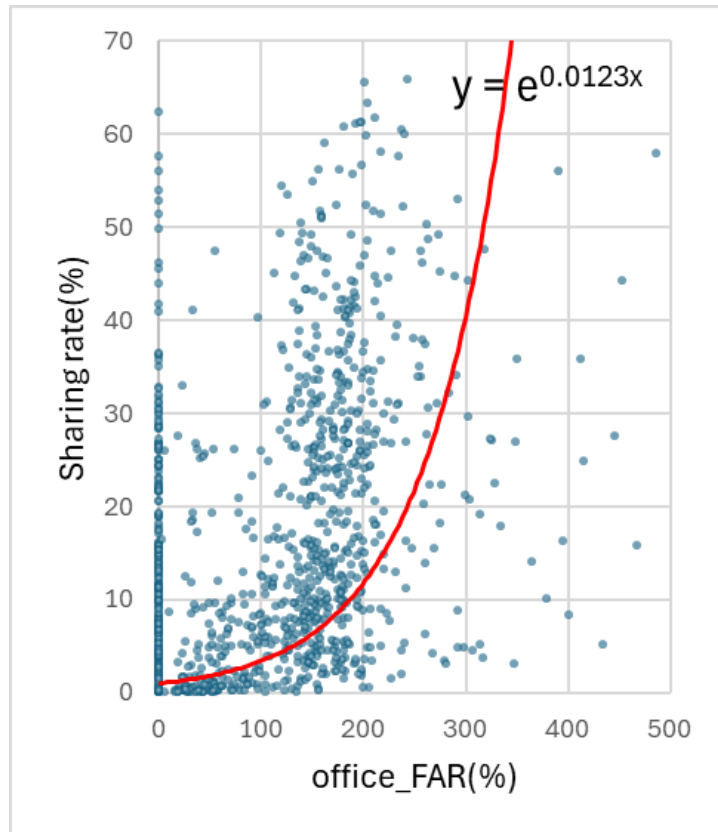
- ✓ Both variables were able to confirm negative relationships, and the influence of the distance variable was greater.



Interpretation of relationships based on XAI

■ Analysis of relationships between key variables (positive effect)

- ✓ It was confirmed that the floor area ratio of the office area had the greatest impact, followed by residential and public areas.

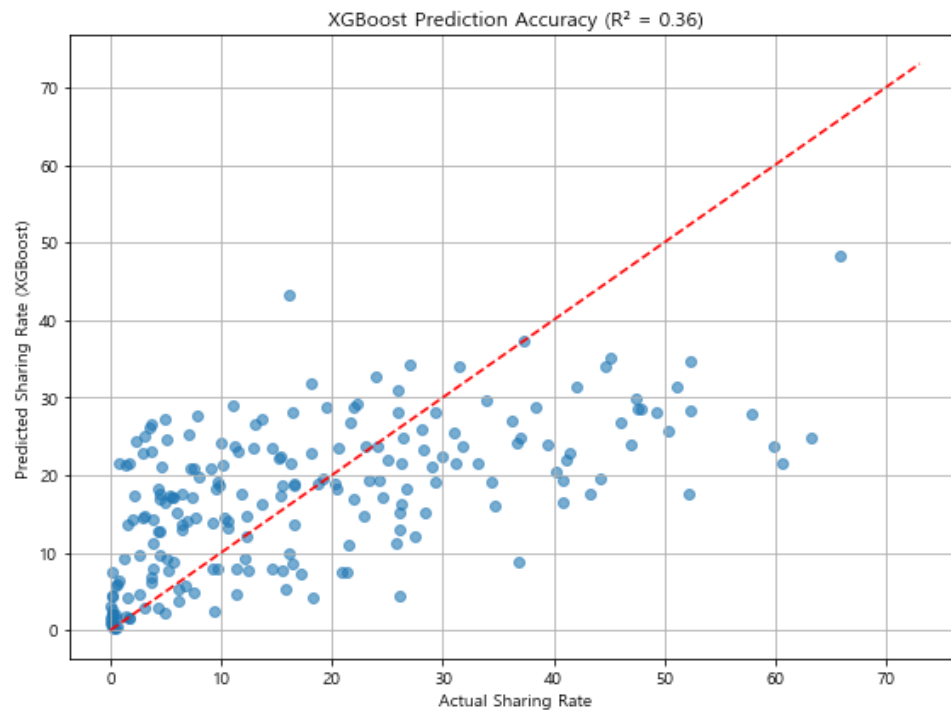


Interpretation of relationships based on XAI

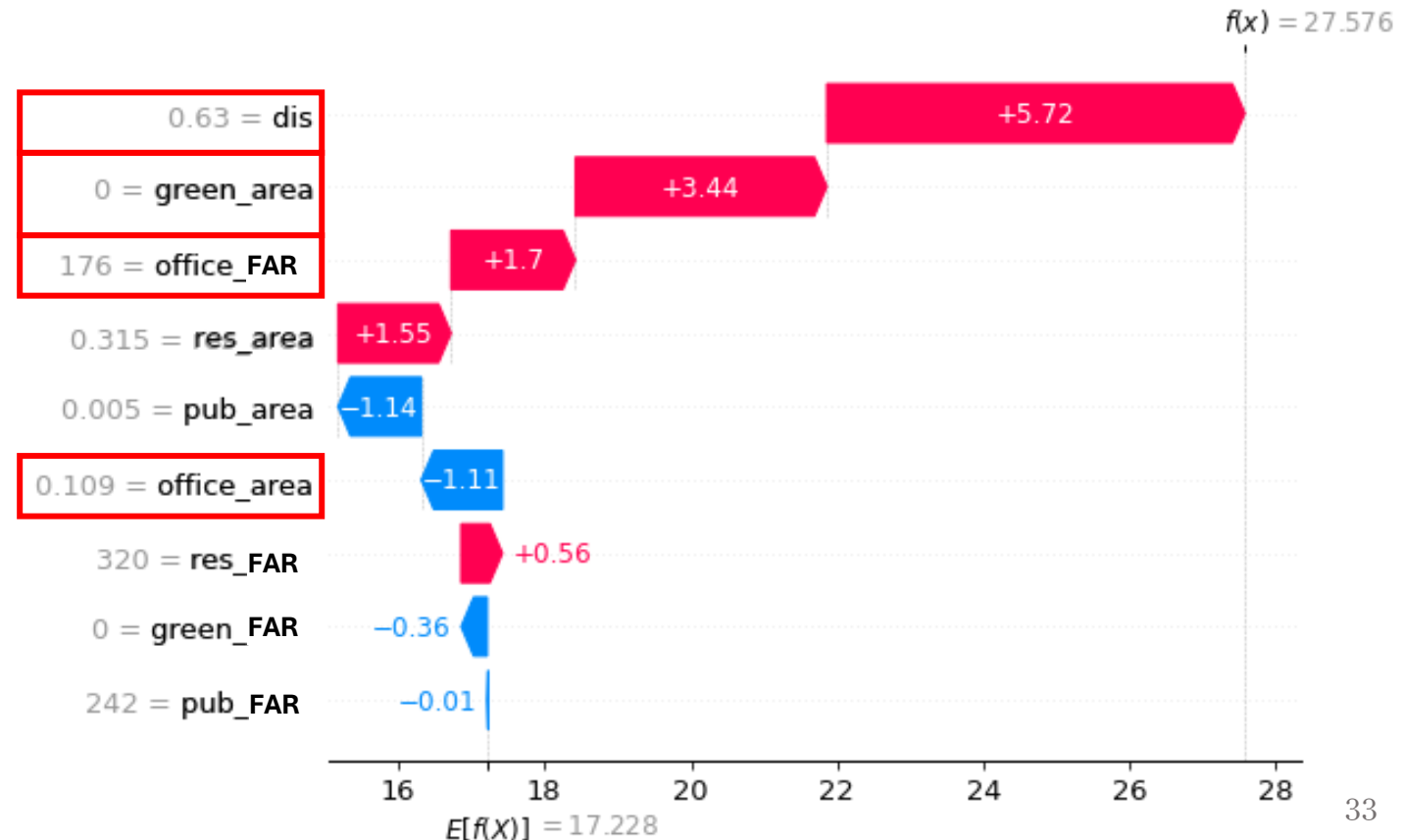
■ A example of Transit Ridership being predicted with AI model

- ✓ This site has a short distance and a small green area, and it has a high FAR value of the office area .
- ✓ Therefore, transit ridership was estimated high.

Validation



Example of a prediction



4. Result of case studies

- I. Experiments on small networks**
- II. Application to the new town area with LLM**

I Experiments on small networks

■ Experiments site

- ✓ To validate the generalization of the proposed framework, case study sites are selected
- ✓ The station of the 1st, 2nd new town and the planned area for future development

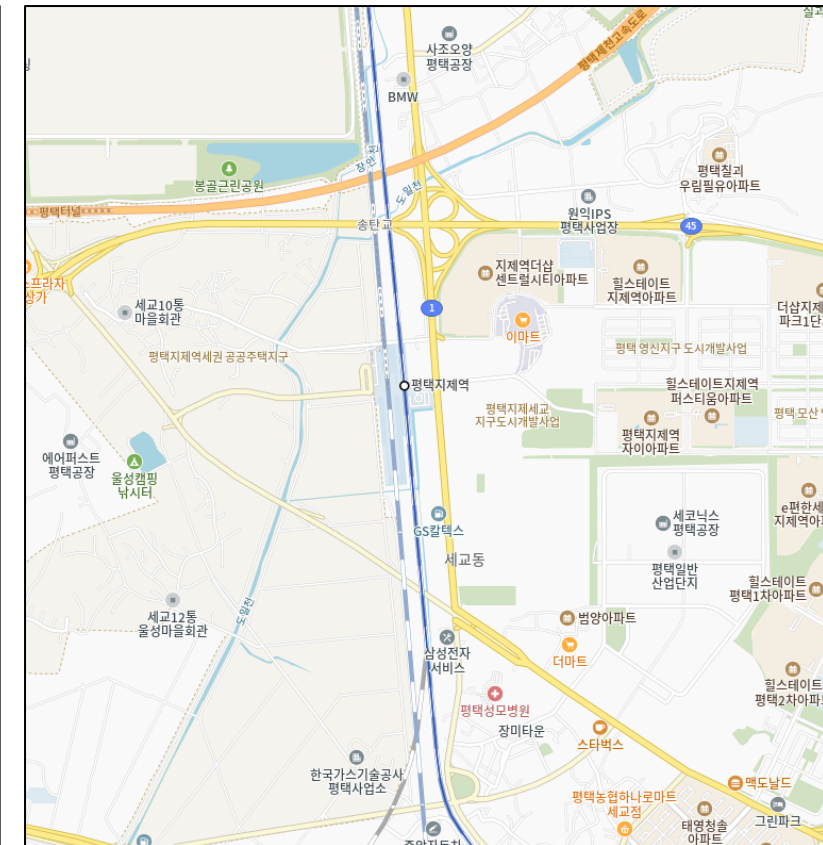
Jeongja station (1st New town)



Macheon station (2nd New town)



PyeongtaekJije station (GTX-A Line)



I Experiments on small networks

■ Experiments site

- ✓ The experimental site was set up as a Jeongja station located in Bundang Newtown.
- ✓ The station has two urban railway lines.



The number of subway users (people/day)

Division		2019	2020	2021	2022	2023
Jeongja	Suin-Bundang Line	13,948	9,066	9,825	10,675	11,208
	Shinbundang Line	7,847	5,626	5,790	6,842	7,627
	Sum	21,795	14,692	15,615	17,517	18,835

Population and transit ridership ratio in the area

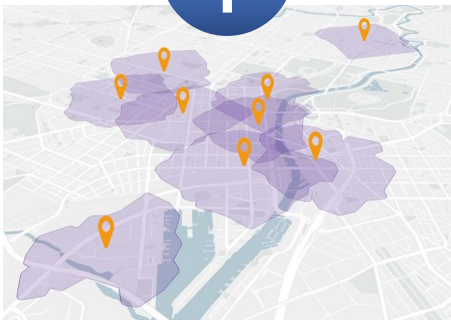
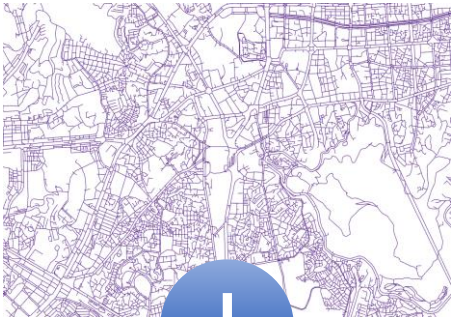
Division	Jeongja dong	Jeongja 1dong	Sum
Population(people)	18,379	28,637	47,016
Transit ridership ratio(%)	17.18	25.84	23.02

I Experiments on small networks

■ Network based catchment area setting

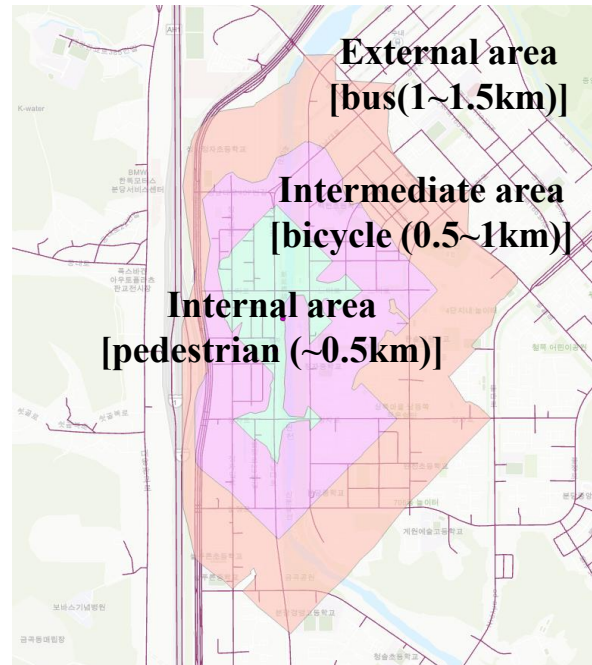
- ✓ Set the Catchment area based on the network distance
- ✓ Separate the area into cells

Road network



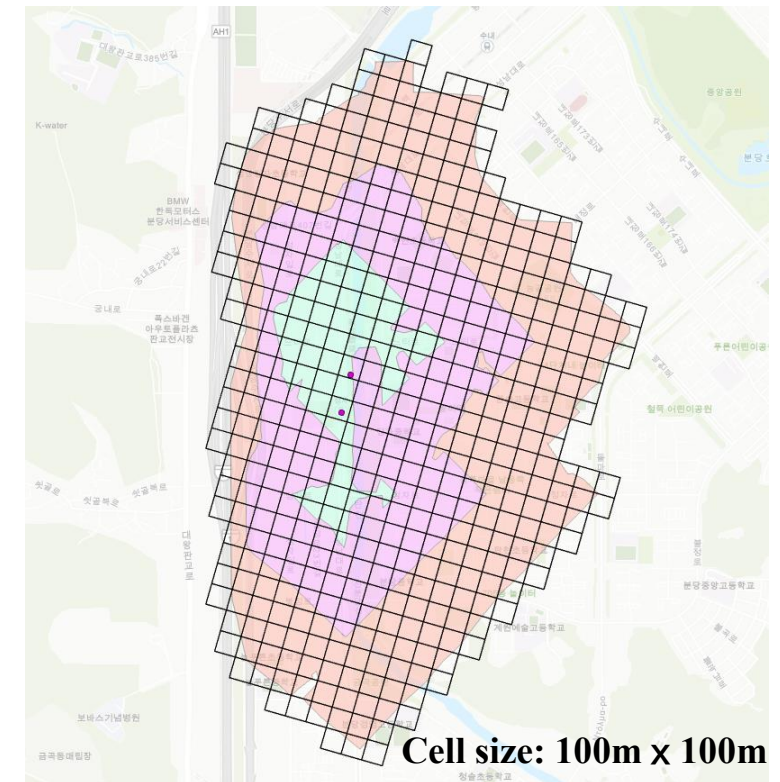
Transit station

Catchment area



Division	Mode	Average speed (km/h)	Travel distance (km)
Internal area	pedestrian	5	0.5
Intermediate area	bicycle	10	1
External area	bus	15	1.5

Separating cells inside the catchment area

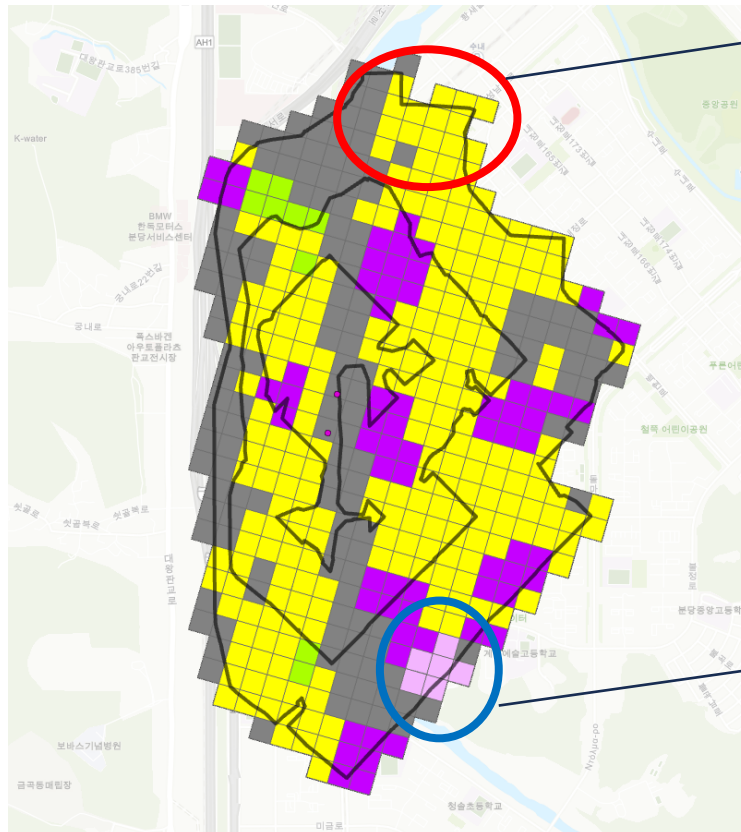


I Experiments on small networks

■ Current status of land use and FAR

- ✓ Considering the relationship of the variables analyzed earlier, this arrangement needs to be improved to enhance the transit ridership

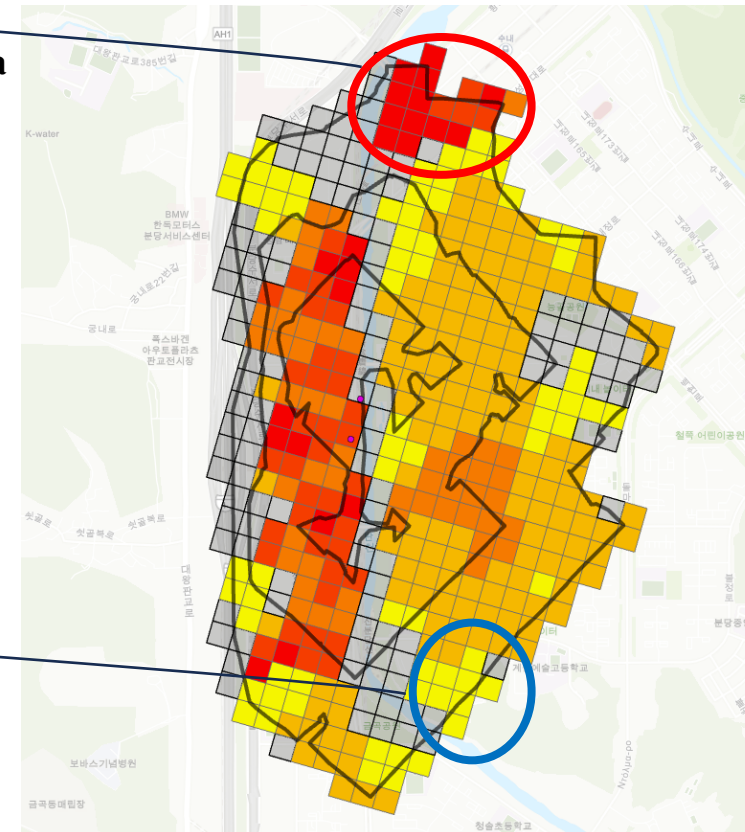
Land use distribution (current)



External area
&
High FAR

Office area
&
Low FAR

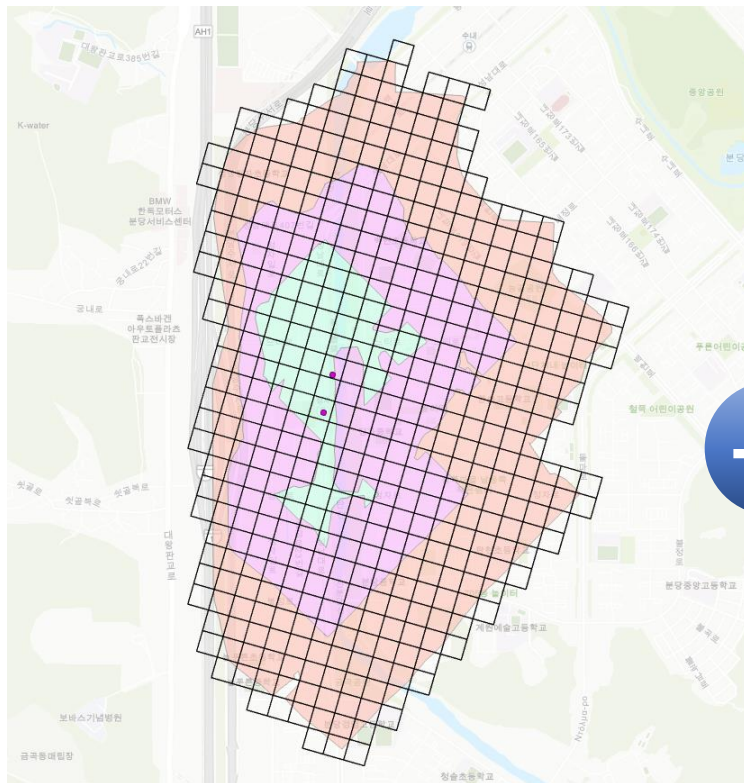
FAR distribution (current)



I Experiments on small networks

■ Generation of land use scenarios

- ✓ land-use type and floor area ratio values are assigned to each cell generated.
- ✓ This process creates various land-use deployment scenarios.



Types of land use

Residential area

Office area

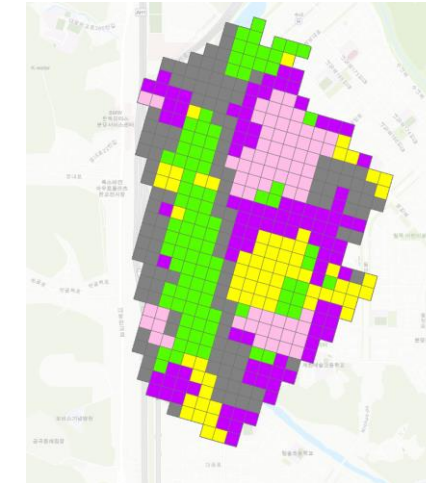
Public area

Green area

Floor area ratio

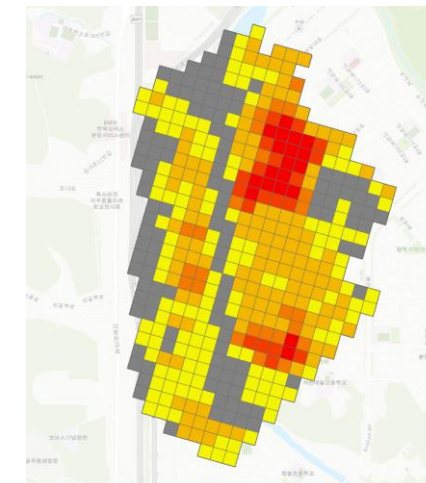
80~350%

Land use type



- Residential area
- Office area
- Public area
- Green area

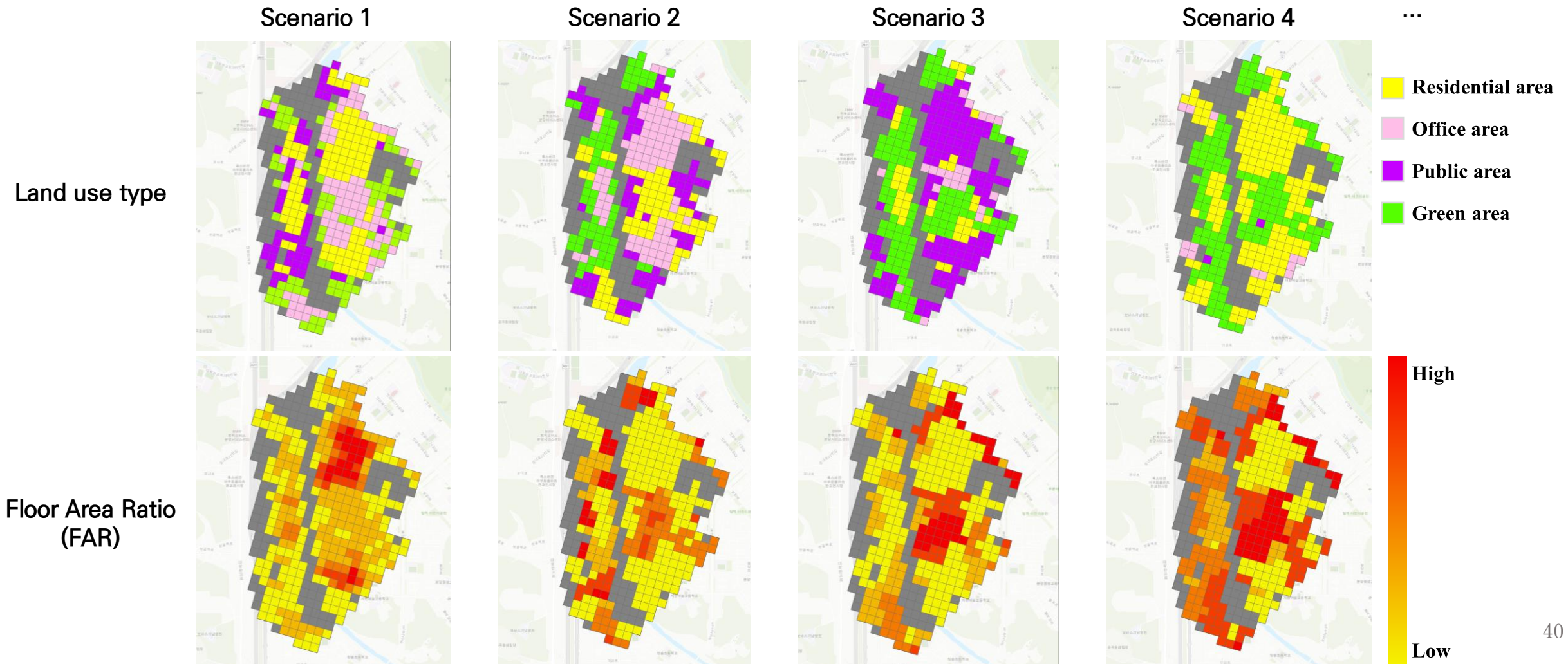
Floor Area Ratio (FAR)



I Experiments on small networks

■ Generation of land use scenarios

- ✓ Random scenarios can be generated in various ways by allocating land-use types and floor area ratios to each cell.



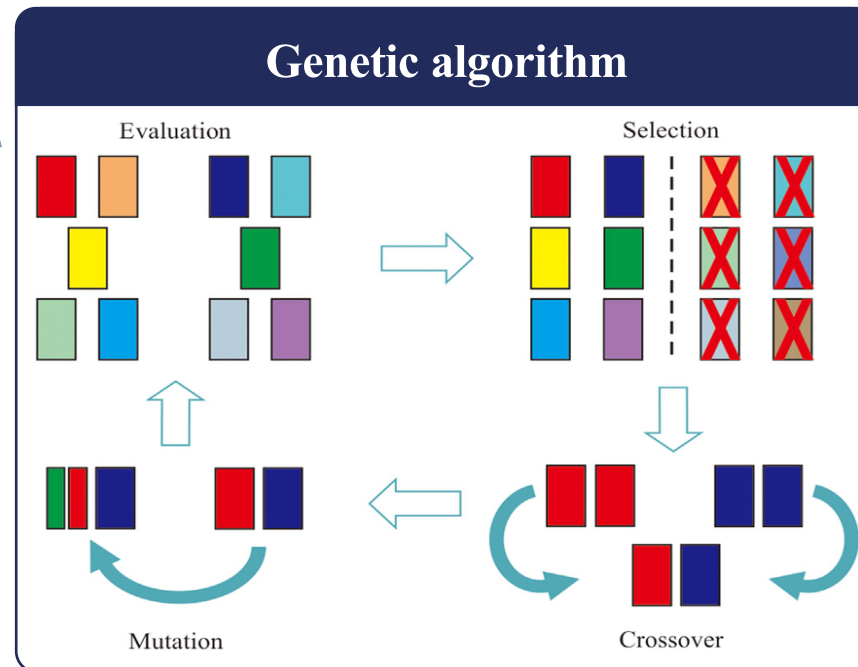
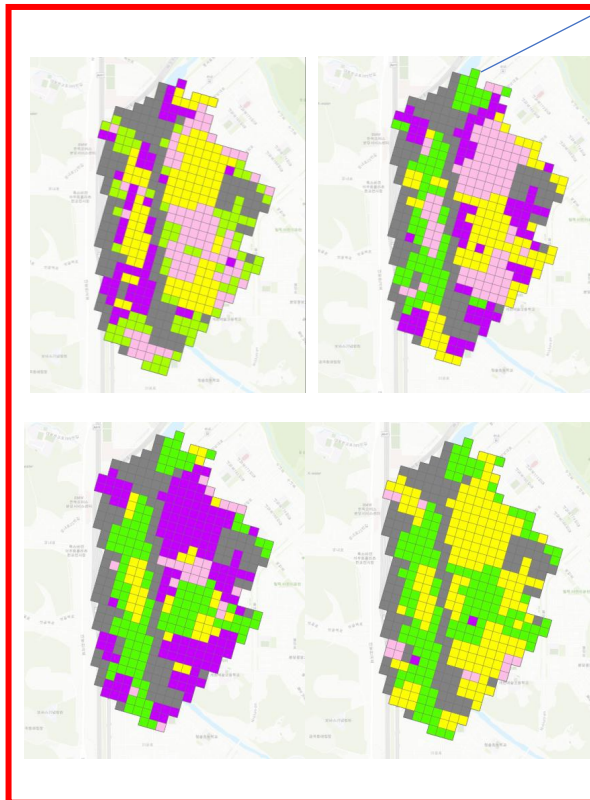
I Experiments on small networks

■ Optimizing scenarios based on genetic algorithms

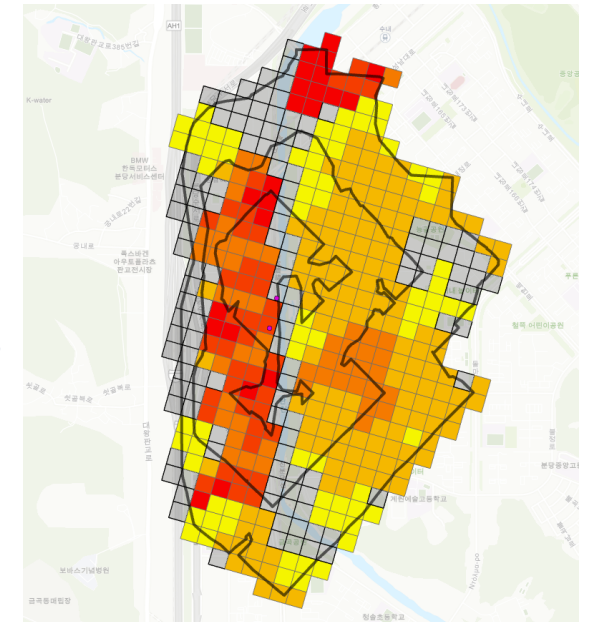
- ✓ Many of these generated scenarios are applied to a genetic algorithm-based optimization process.
- ✓ The entire grid is defined as one chromosome, and each cell acts as a gene.

Chromosomes

■ Gene = [land use type, FAR]



Optimal solution



Fitness: 3E performance indicators

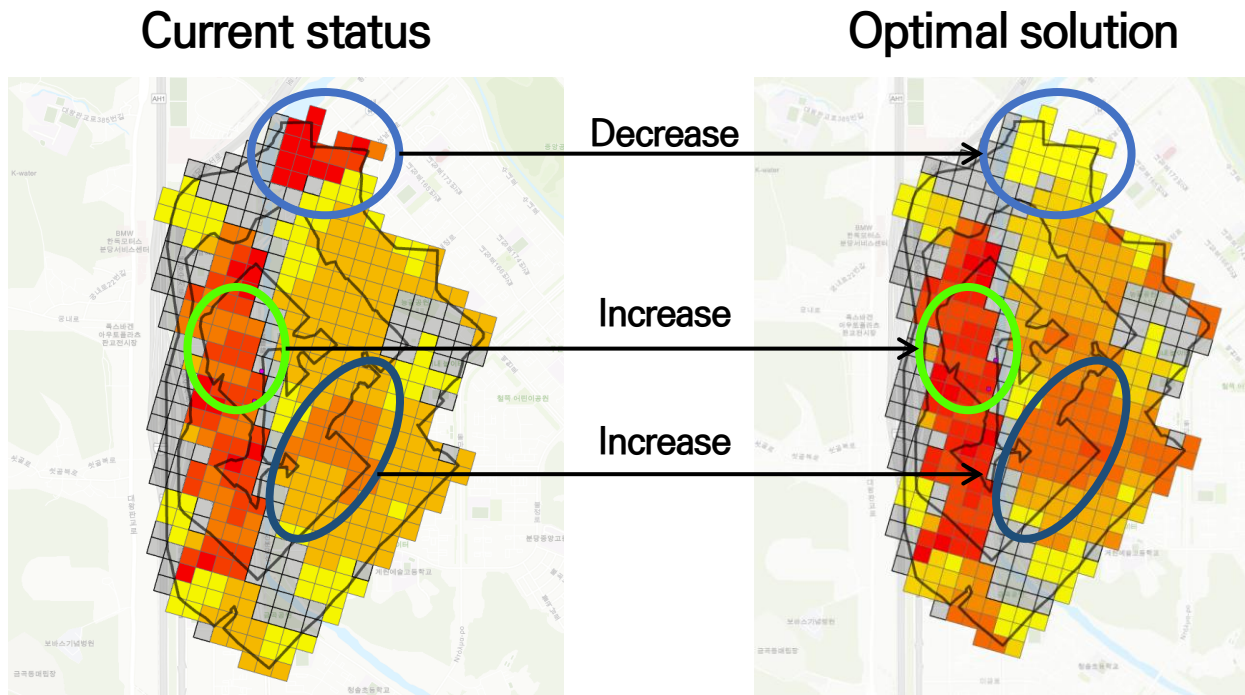
$$\text{Max}(W_1 \text{ Transit ridership} + W_2 \text{ Accessibility} + W_3 \text{ Transportation Emission})$$

Division	W_1	W_2	W_3
Basic plan	0.33	0.33	0.33

I Experiments on small networks

■ Optimizing scenarios based on genetic algorithms

- ✓ Compared to the existing state, structural improvements are identified.
- ✓ As a result, the transit ridership ratio increased about 5 percentage points



Change of the population and transit ridership ratio

Division	Before	After	Difference
Population(people)	47,016	56,420	9,404
Transit ridership ratio(%)	23.02	27.82	4.8

Distribution of the population

Division	Population	Ratio
Internal area(pedestrian)	24,836	44.02%
Intermediate area(bicycle)	19,156	33.95%
External area(bus)	12,429	22.03%
Sum	56,420	100%

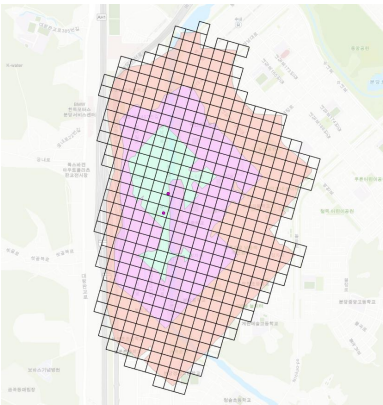
I Experiments on small networks

■ Case Studies in New town and empty city

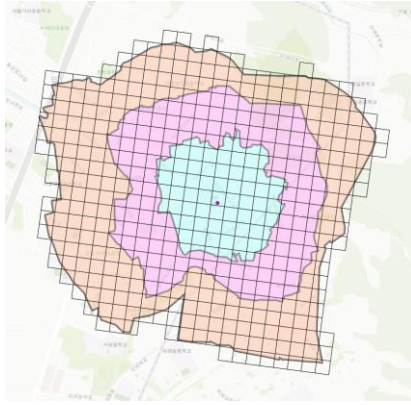
- ✓ To validate the generalization of the proposed framework, additional case study sites are selected
- ✓ The station of the 2nd new town and the planned area for future development

Case study site

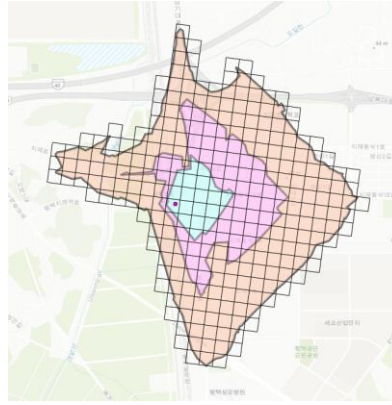
Jeongja station
(1st New town)



Macheon station
(2nd New town)



PyeongtaekJije station
(GTX-A Line)



Comparison of the performance

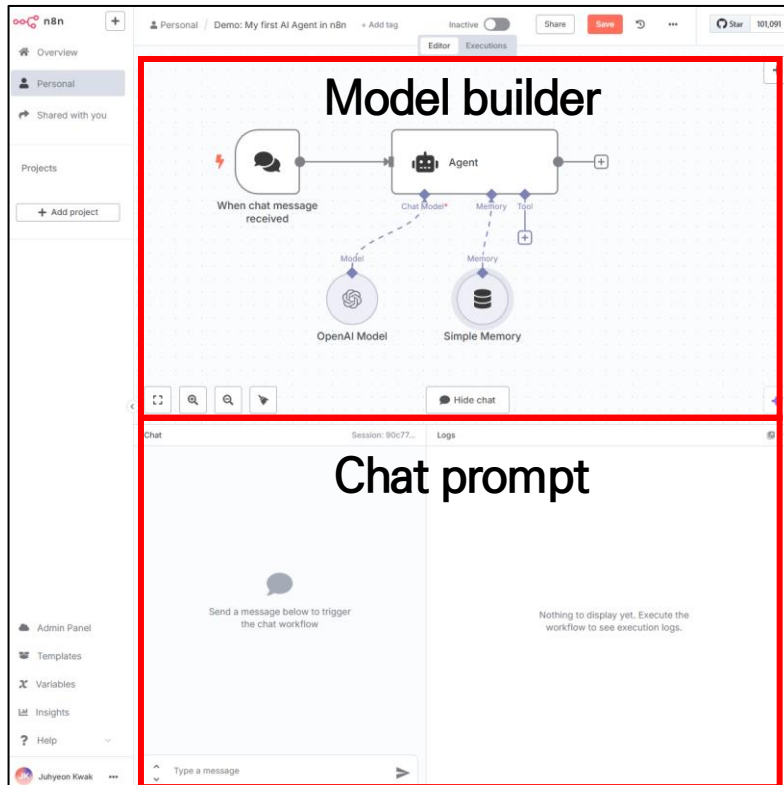
Station	Transit Ridership	Accessibility	Transportation emission
Jeongja	+4.80%p	+3.12%p	-0.91%p
Macheon	+6.40%p	+7.28%p	-1.49%p
PyeongtaekJije	+10.63%p	+8.15%p	-2.17%p

II Application to the new town area with LLM

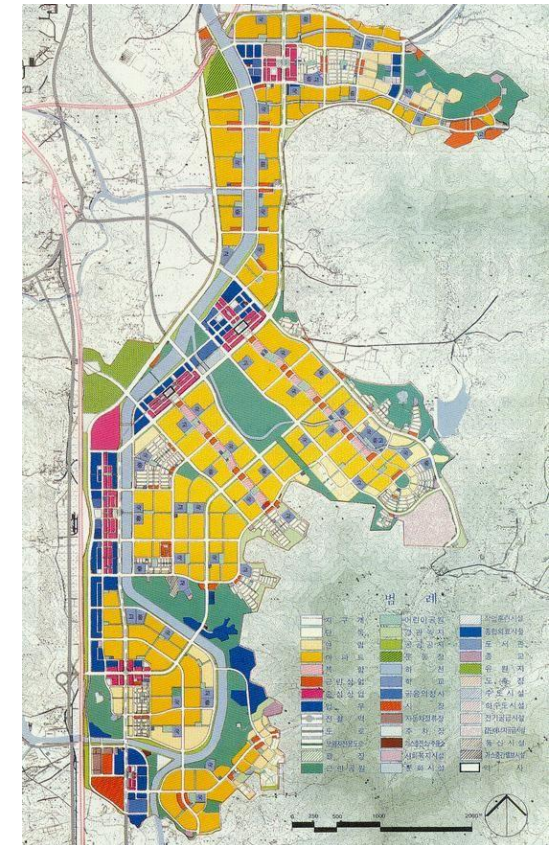
4. Result of case studies

■ Introduction of LLM agent

- ✓ An AI agent that receives analyzed information in memory is introduced.
- ✓ And the case study site is extended to all of the Bundang Newtown



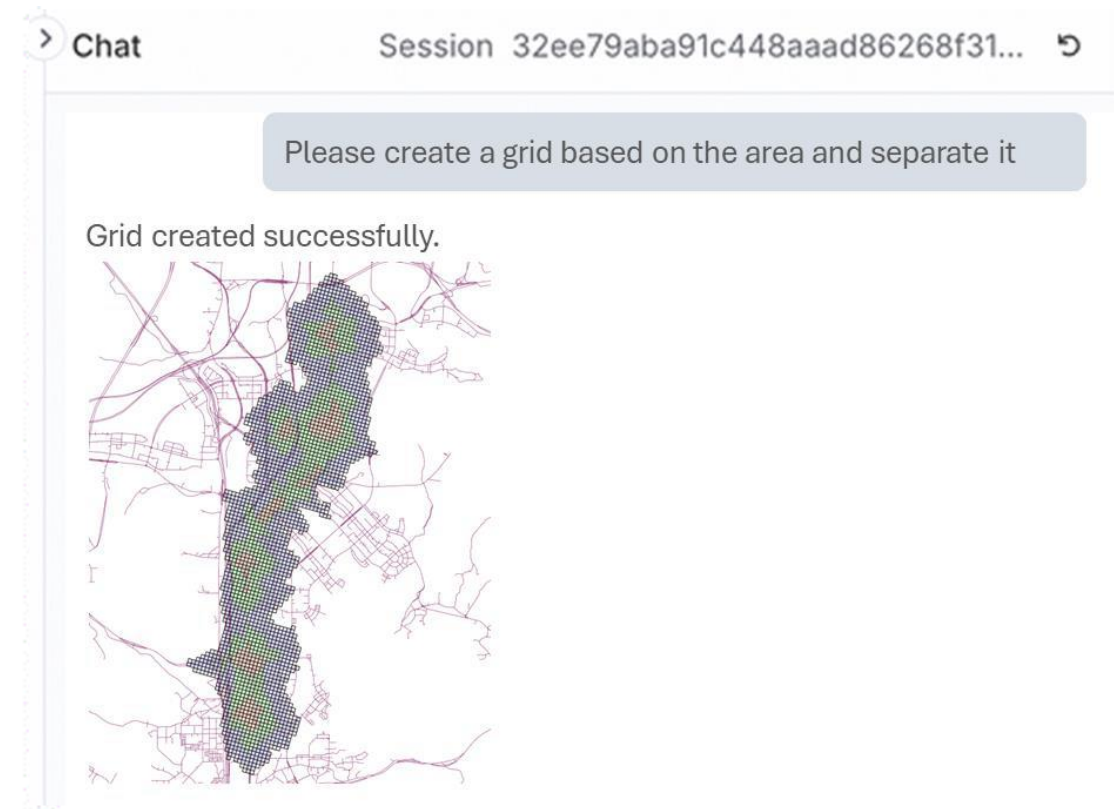
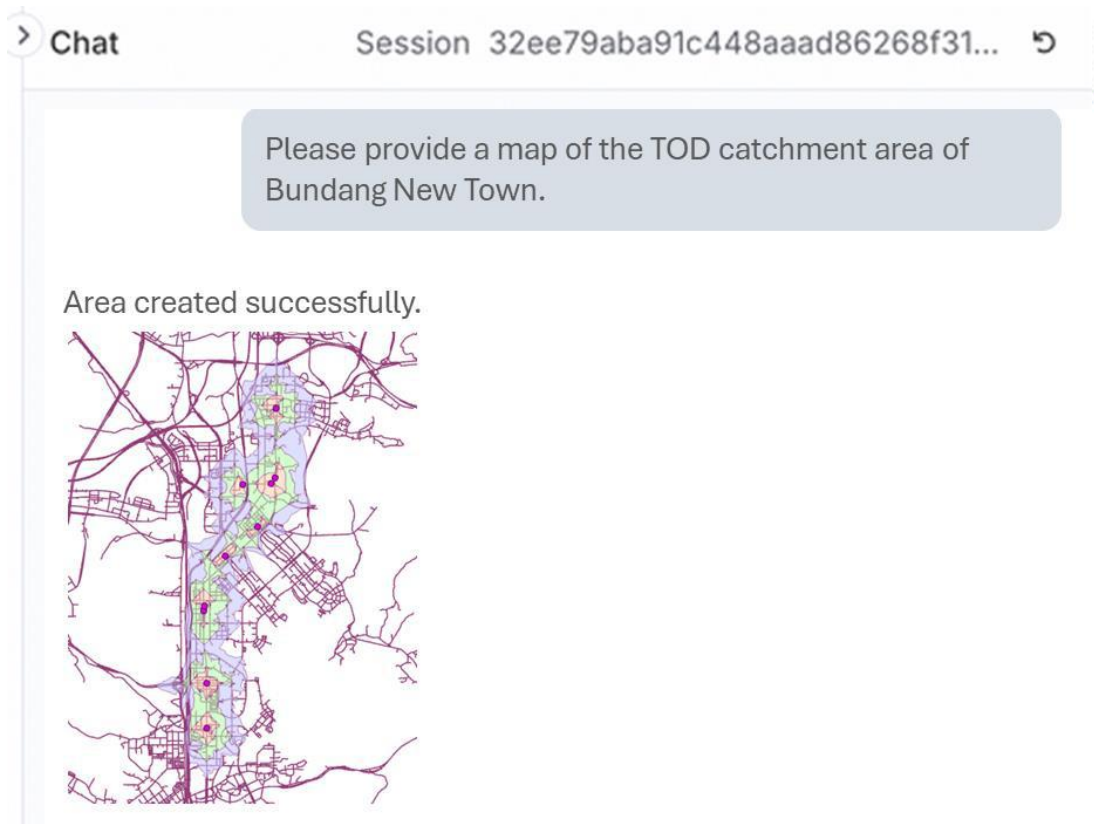
Bundang Newtown



II Application to the new town area with LLM

■ Network based catchment area setting with LLM

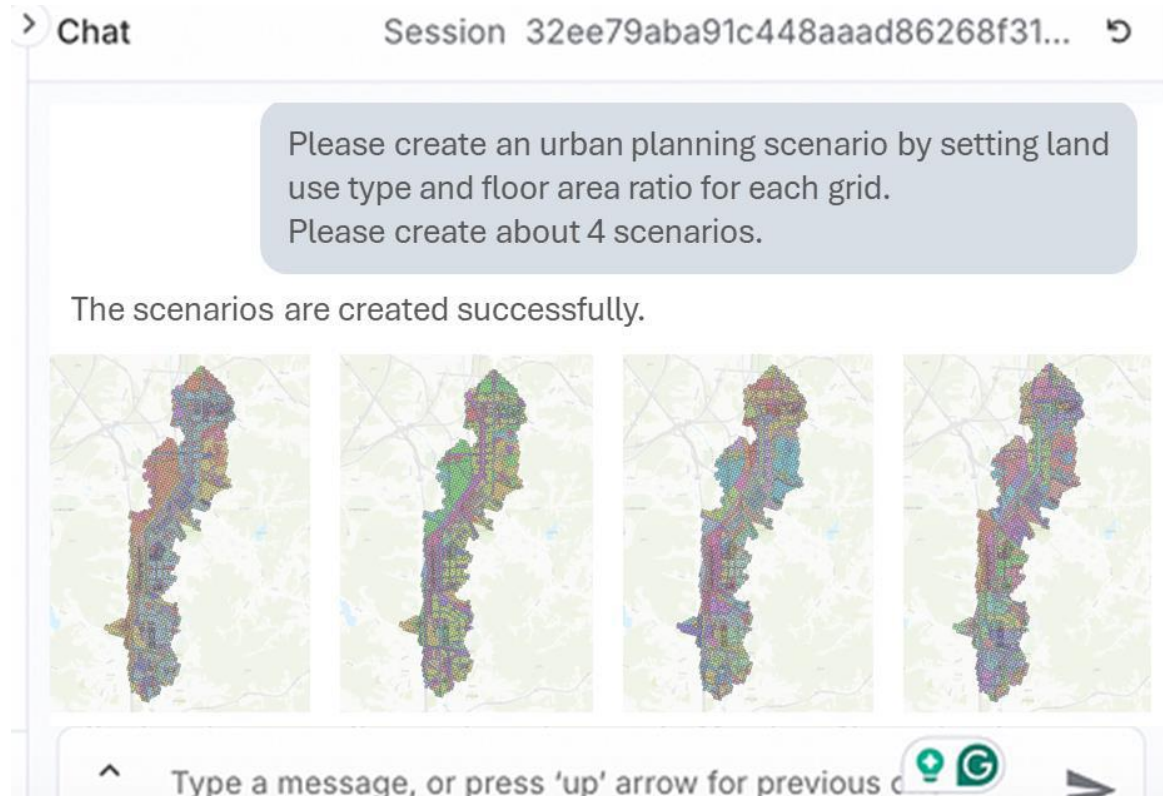
- ✓ An AI agent that receives analyzed information in memory is introduced.
- ✓ And the case study site is extended to all of the Bundang Newtown



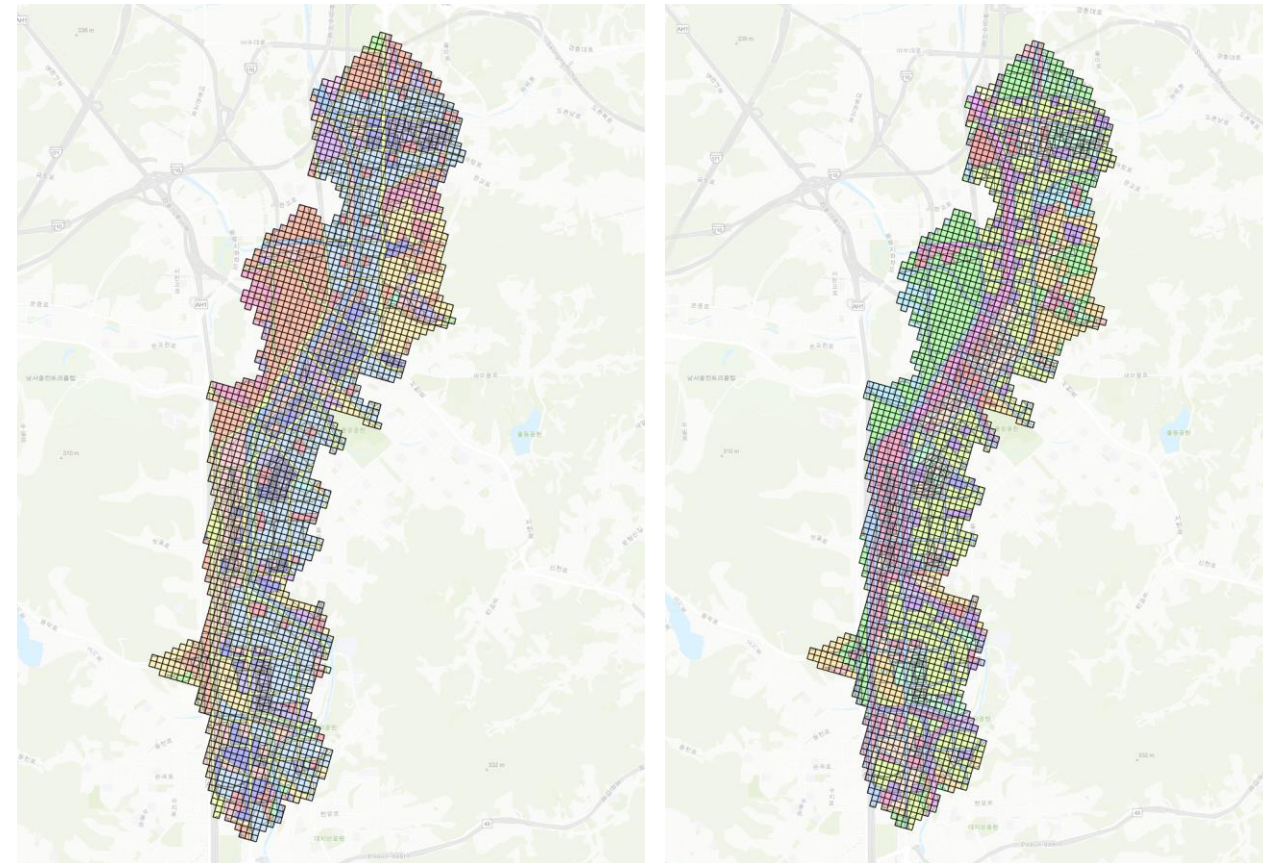
II Application to the new town area with LLM

■ Generation of Scenarios with LLM

- ✓ AI agents automatically generate scenarios by combining pre-trained land-use types and floor area ratios



Samples of the scenario



II Application to the new town area with LLM

■ Optimizing scenarios based on genetic algorithms

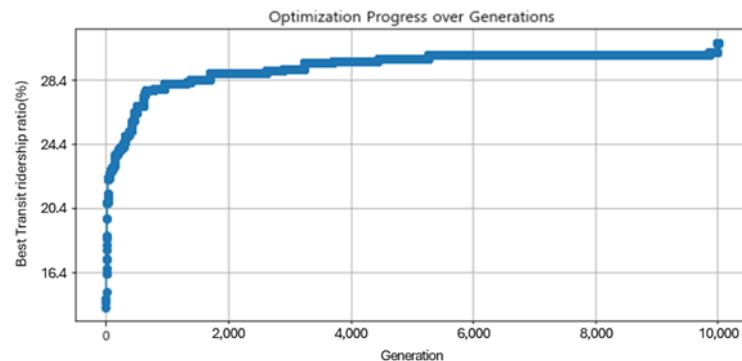
- ✓ Derive optimal solutions based on genetic algorithms

> Chat

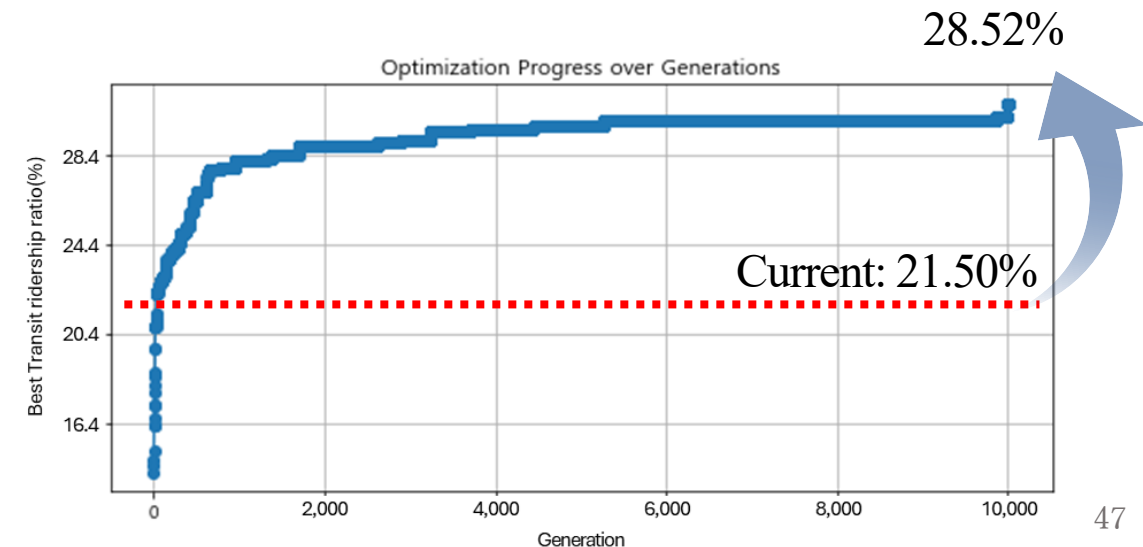
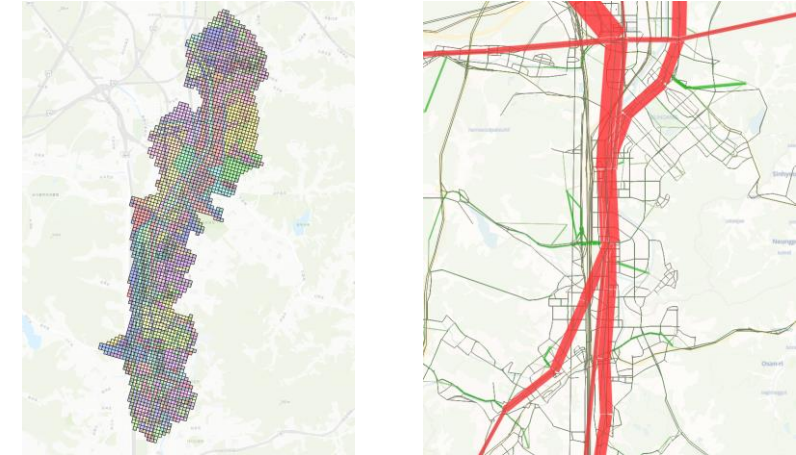
Session 32ee79aba91c448aad86268f31...

Use genetic algorithms to derive optimal solutions. Chromosomes use the previously generated scenario, and Fitness uses the output value of the AI-based Transit ridership model.

The optimal solution has been derived.
The transit ridership ratio converged to 28.52%.
Below is the learning process and optimal solution map.



Optimal solution and transit assignment



5. Conclusion

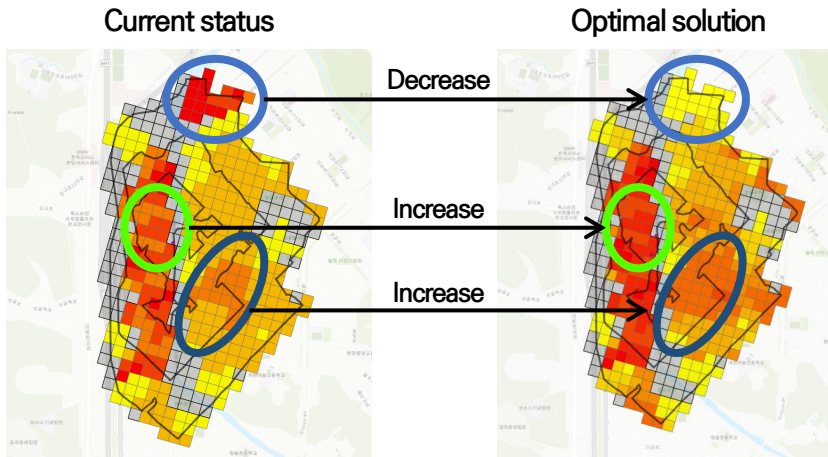
- I. Conclusion
- II. Discussion: Strengths of LLM in planning
- III. Expected effects

I Conclusion

■ Summary of the research objective

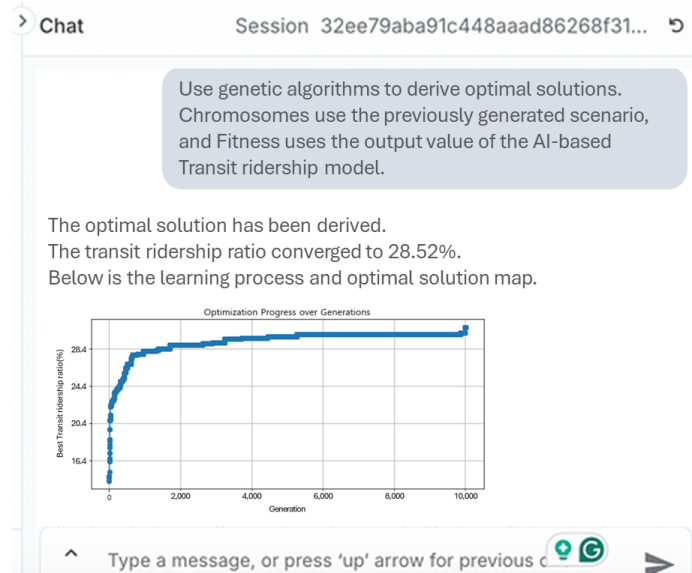
I

Finding the optimal land use layout to maximize TOD performance



II

Support decision making with LLM-based TOD design model



III

Improve network efficiency with optimal TOD design to respond to climate change

Station	Transit Ridership	Accessibility	Transportation emission
Jeongja	+4.80%p	+3.12%p	-0.91%p
Macheon	+6.40%p	+7.28%p	-1.49%p
Pyeongtaek-Jije	+10.63%p	+8.15%p	-2.17%p

II Discussion: Strengths of LLM in planning

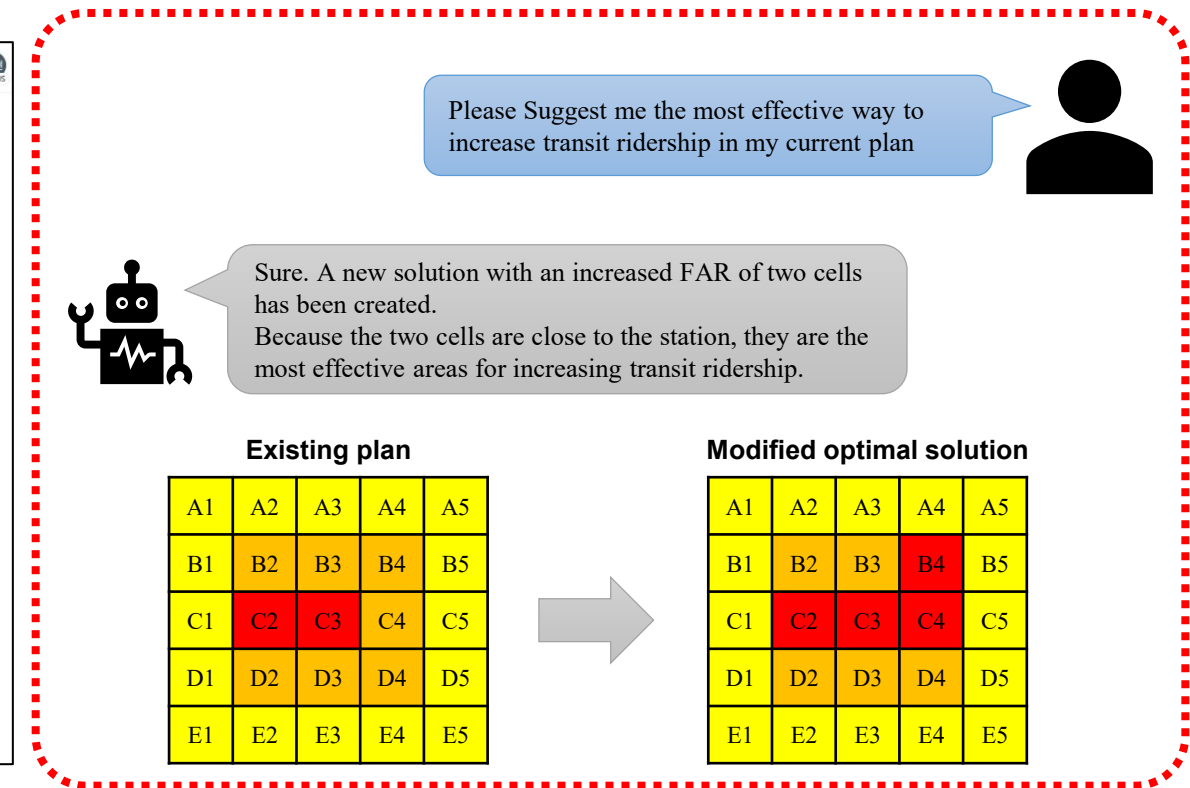
■ Strengths of LLM in planning

- ✓ The planning process is difficult to accommodate all the needs because the needs of various stakeholders are intertwined
- ✓ When LLM is introduced into the planning process, we can just judge stakeholders' opinions and check the revised solution.

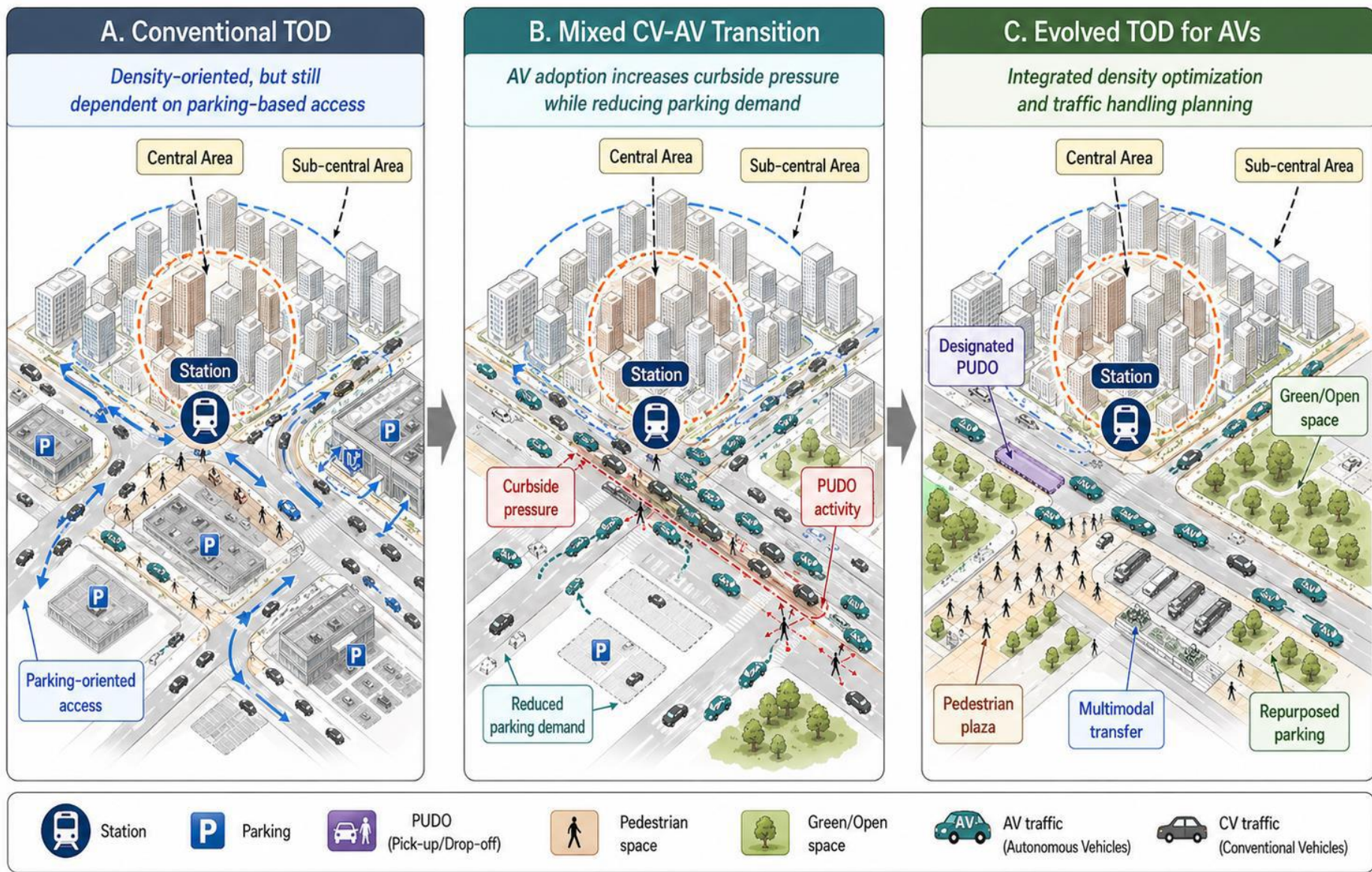
Modifying the results of ChatGPT



Strengths of LLM in the planning process



II Discussion: Evolution of TOD in the AV era



III Expected effects

■ The expected effects of the study



Academic

Developing methodology for TOD design

- Improving network efficiency by optimizing land use layout
- Improving the reliability of LLMs for TOD design by validating based on real data



Economic

Saving the social costs

- Reducing congestion and air pollution costs by improving network efficiency
- Increasing the value of land use through optimized urban transportation design



Social

Improve the social equity

- Promoting transportation welfare by improving the accessibility of public transportation
- Contributing to net-zero achievement by reducing carbon emissions in the transport sector



Supporting decision-making with Large Language Models-based TOD design





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Thank you